Your house: light, sun, heat and wind

Casa Maringá · simulation report · Maringá, PR (lat -23.41, lon -51.93) · 17 October 2026
Weather: typical year from BRA_PR_Maringa.868990_TMYx.2011-2025.epw · annual sun 1,882 kWh/m² · mean temperature 23.0 °C · mean wind 2.3 m/s

How to read this report. Every number comes with a coloured tag telling you whether it is good or bad, on a **level from 1 (bad) to 5 (excellent)**:

Excellent 5/5 Good 4/5 Acceptable 3/5 Poor 2/5 Bad 1/5

Each tag is based on a **Brazilian norm (ABNT)** where one exists, on an **international standard** where Brazil gives no number, or on a clearly labelled **rule of thumb**. The boxes *What counts as good?* at the end of each section show the full scale and the source. The model walls are assumed to be: *Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)*; roof: *Clay tile + ventilated attic + 12 cm concrete slab ceiling*. These materials and the window sizes (common catalogue sizes of about 2012) are estimates, not measurements: use the results to compare rooms and options, not as a guarantee.

1. The short version

Each room is rated on up to nine checks, each on a **level from 1 (bad) to 5 (excellent)**; the OVERALL column is the average level. The next pages explain every check; the cooling column shows what an air conditioner would cost per year:

Room	Daylight (NBR 15575)	Lamps off in daytime	Sun on window	Comfort hours	Hot day	Cold day	Openable window	Cooling cost	OVERALL
Living + dining	Intermediate 4/5	Often 4/5	Below minimum 2/5	Good 4/5	Intermediate 4/5	Superior 5/5	Meets minimum 3/5	R$ 844/yr Low 4/5	Good 3.7/5
Hallway	No requirement 3/5	Almost never 1/5	-	Excellent 5/5	-	-	-	-	Acceptable 3.0/5
Bedroom 1	Superior 5/5	Almost always 5/5	Medium 4/5	Good 4/5	Superior 5/5	Superior 5/5	Fails 1/5	R$ 153/yr Very low 5/5	Good 4.1/5
Bathroom 1	Intermediate 4/5	Almost never 1/5	-	Good 4/5	-	-	-	-	Acceptable 3.0/5
Bedroom 2	Intermediate 4/5	Sometimes 3/5	Below minimum 2/5	Good 4/5	Superior 5/5	Superior 5/5	Fails 1/5	R$ 112/yr Very low 5/5	Good 3.6/5
Bathroom 2	Superior 5/5	Almost never 1/5	-	Good 4/5	-	-	-	-	Acceptable 3.3/5
Bedroom 3	Superior 5/5	Often 4/5	Below minimum 2/5	Good 4/5	Superior 5/5	Superior 5/5	Fails 1/5	R$ 110/yr Very low 5/5	Good 3.8/5
Kitchen	Superior 5/5	Almost always 5/5	Below minimum 2/5	Acceptable 3/5	-	-	Meets minimum 3/5	R$ 363/yr Moderate 3/5	Good 3.7/5
Outdoor bath	-	Rarely 2/5	-	Acceptable 3/5	-	-	-	-	Acceptable 2.5/5
Outdoor storage	-	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.0/5
Whole house								**R$ 1,582/yr**	Good 3.6/5

What works well

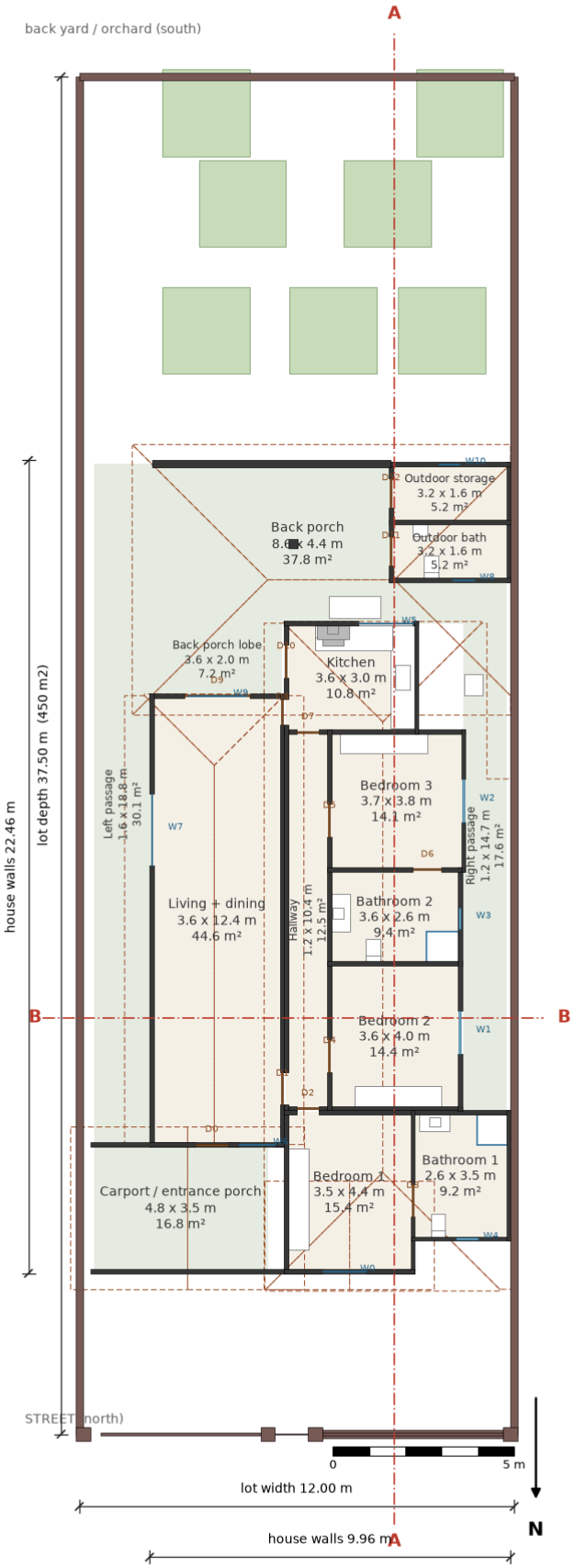
- TV / dining (rear half): 408 lux of daylight at the centre (NBR 15575 superior level)
- Bedroom 1: 513 lux of daylight at the centre (NBR 15575 superior level)
- Bedroom 3: 123 lux of daylight at the centre (NBR 15575 superior level)
- Kitchen: 188 lux of daylight at the centre (NBR 15575 superior level)
- Most comfortable room: Hallway (92% of the year in the comfort band; outdoors only 48%)
- External wall heat transmittance U: 2.52 W/m²K meets the limit <= 3.7
- External wall heat capacity: 134.0 kJ/m²K meets the limit >= 130

What deserves attention

- Living (front half): lamps are needed for about 53% of the daytime hours
- Bedroom 2: lamps are needed for about 60% of the daytime hours
- Least comfortable room: Outdoor bath (73%; 1546 h too hot, 815 h too cold)
- Bedroom 1: openable window area only 4.7% of the floor (NBR 15575-4 asks 7%)
- Bedroom 2: openable window area only 5.0% of the floor (NBR 15575-4 asks 7%)
- Bedroom 3: openable window area only 5.1% of the floor (NBR 15575-4 asks 7%)
- Roof heat transmittance U (indicative): 1.78 W/m²K vs limit <= 1.5

FLOOR PLAN - ground floor

House faces NORTH (street at the bottom) - rooms with name, size and area; sections A-A a



Scale 1:150 - sizes are estimates from a video scan and the listing, not a survey

Floor plan of the model (north at the bottom).

2. Natural light

Daylight saves electricity and makes rooms pleasant. The Brazilian norm NBR 15575 asks for at least **60 lux** at the centre of living rooms, bedrooms and kitchens (lux = how much light lands on a table). For comparison: a bright overcast day outdoors is thousands of lux.

2.1 Brazilian test: light at the centre of the room, lamps off

Room	Lux at centre	Verdict
Living (front half)	91	Intermediate 4/5
TV / dining (rear half)	408	Superior 5/5
Hallway	0	No requirement 3/5
Bedroom 1	513	Superior 5/5
Bathroom 1	42	Intermediate 4/5
Bedroom 2	93	Intermediate 4/5
Bathroom 2	48	Superior 5/5
Bedroom 3	123	Superior 5/5
Kitchen	188	Superior 5/5
Outdoor bath	53	Fails 1/5
Outdoor storage	139	Superior 5/5

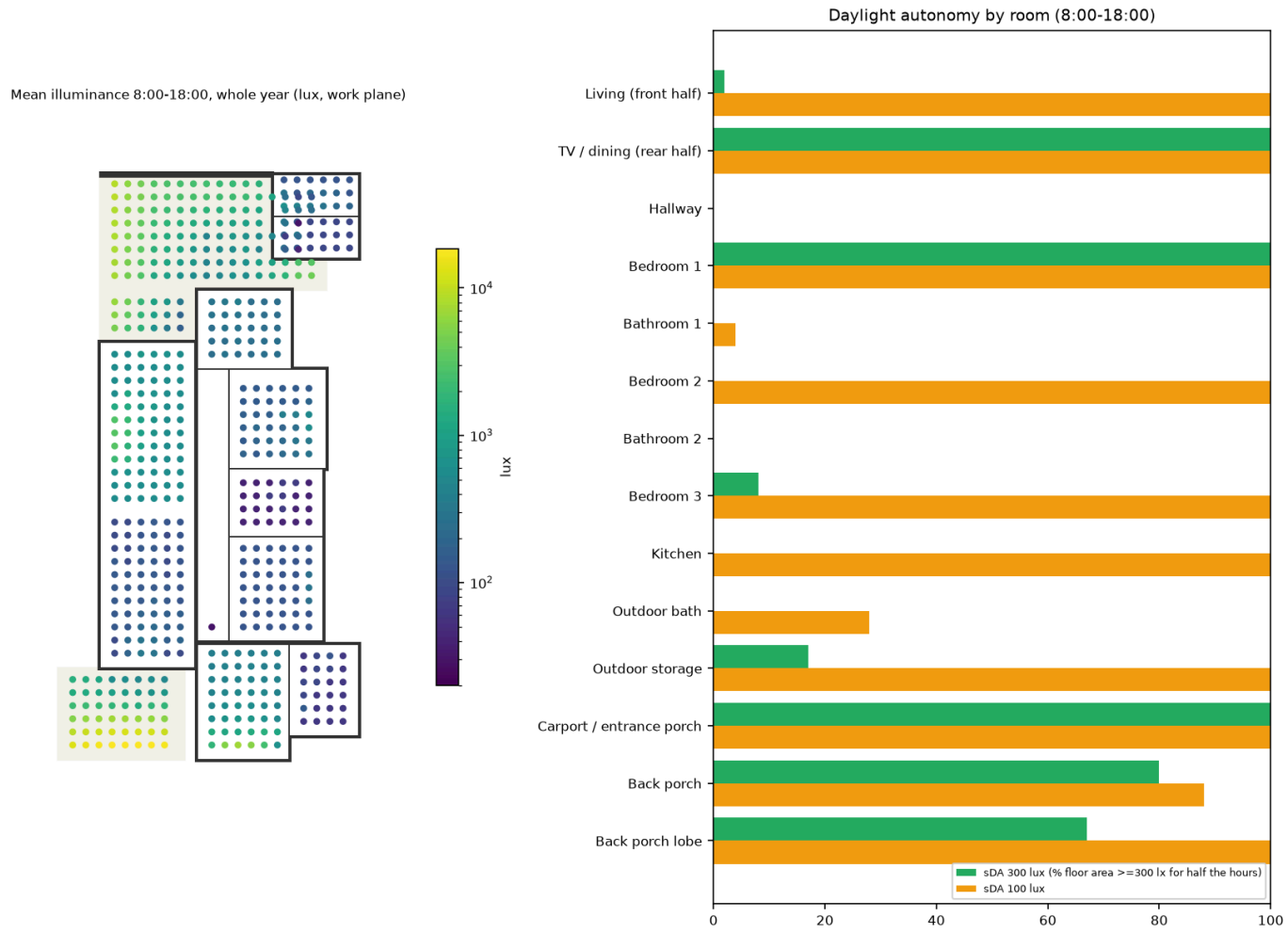
Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

2.2 How much of the room is bright, and when do I need a lamp?

Room	Goal (lux)	Floor at the goal	Daytime with lamps off	Strong sun (ASE)
Living (front half)	150	26% Below target 2/5	47% Sometimes 3/5	0% Very low 5/5
TV / dining (rear half)	150	100% Whole room 5/5	96% Almost always 5/5	0% Very low 5/5
Hallway	-	-	0% Almost never 1/5	0% Very low 5/5
Bedroom 1	100	100% Whole room 5/5	96% Almost always 5/5	10% High 2/5
Bathroom 1	-	-	1% Almost never 1/5	0% Very low 5/5
Bedroom 2	100	100% Whole room 5/5	40% Sometimes 3/5	0% Very low 5/5
Bathroom 2	-	-	0% Almost never 1/5	0% Very low 5/5
Bedroom 3	100	100% Whole room 5/5	70% Often 4/5	0% Very low 5/5
Kitchen	200	100% Whole room 5/5	81% Almost always 5/5	0% Very low 5/5
Outdoor bath	-	-	14% Rarely 2/5	0% Very low 5/5
Outdoor storage	-	-	78% Almost always 5/5	0% Very low 5/5

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

Goal = the light level EN 17037 (UK National Annex for dwellings) recommends from daylight: 100 lux bedroom, 150 lux living room, 200 lux kitchen. 'Lamps off' = share of the hours 8:00-18:00 in which the daylight average already reaches the general light level of section 3.

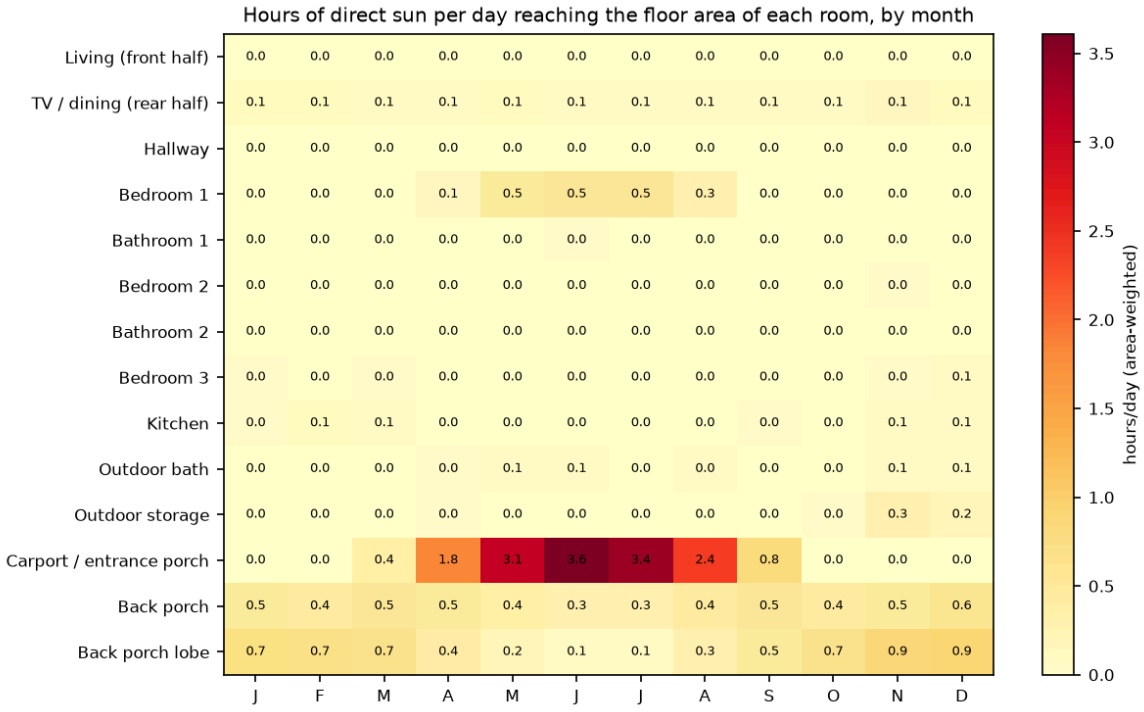


Left: average light on the floor over the year. Right: share of each room that reaches 100 and 300 lux.

2.3 Sun reaching the windows

Room	Sun on window, hours/day (early spring)	Verdict
Living (front half)	0.0	Below minimum 2/5
TV / dining (rear half)	0.3	Below minimum 2/5
Bedroom 1	3.7	Medium 4/5
Bedroom 2	0.0	Below minimum 2/5
Bedroom 3	0.0	Below minimum 2/5
Kitchen	0.0	Below minimum 2/5

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9



Hours per day of direct sun that reach the floor of each room, by month (J = January).



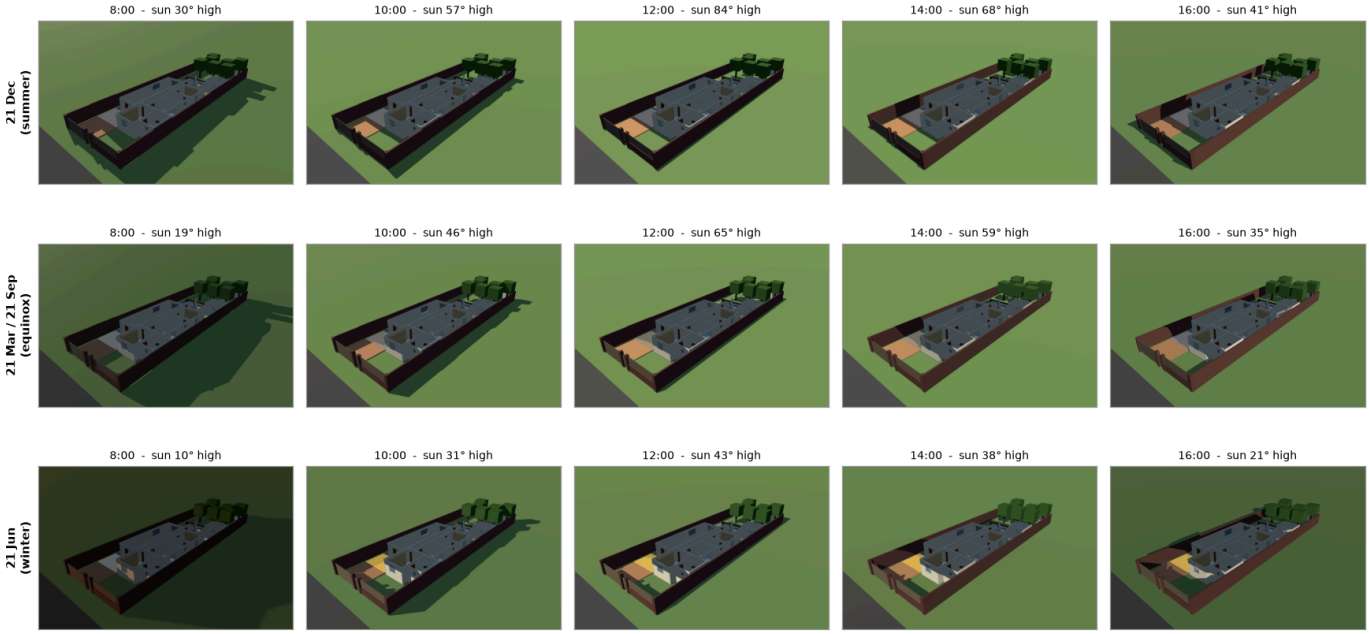
Where the sun lands on the floor on four key dates.

2.4 The sun and its shadows through the day in the 3D model

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

These pictures come from the same 3D model and sun calculation as the interactive viewer: three key dates, five times of day. Shadows of the house, the roof, the boundary walls and the trees are drawn; the floors that receive direct sun turn yellow/orange. in winter the low sun reaches deep into the rooms that face it, while in summer the sun is high at noon and the roof shades the house.

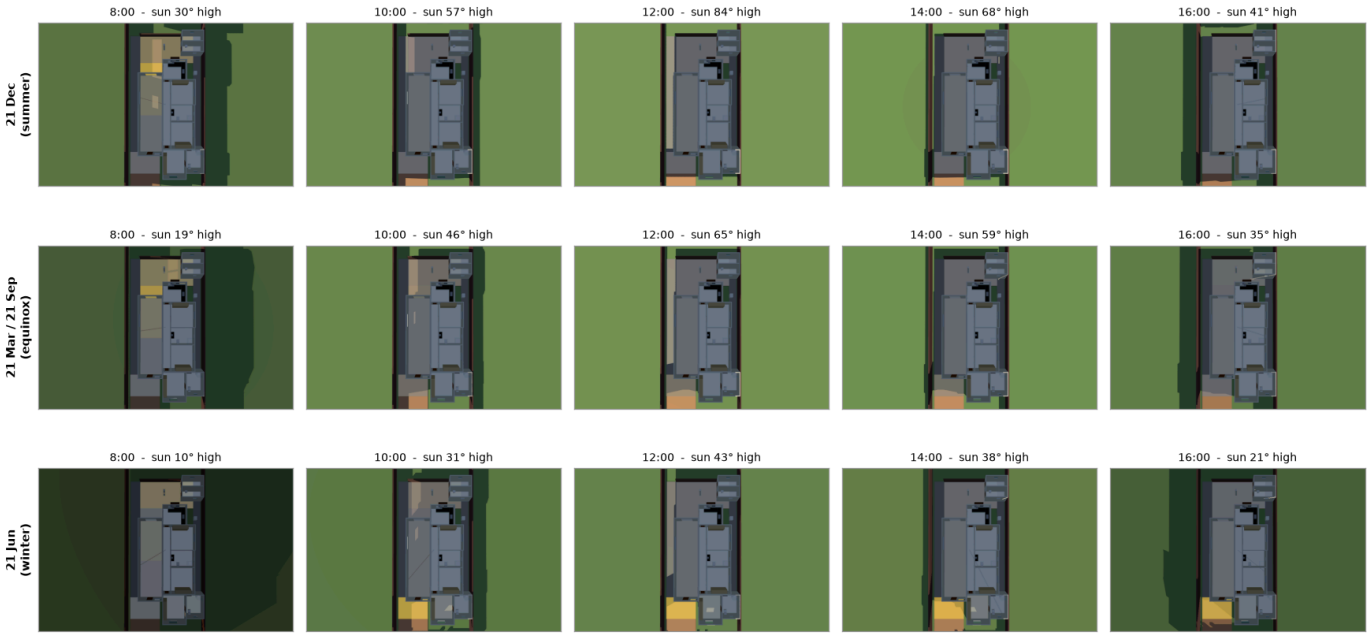
The sun and its shadows through the day on the 3D model (view from the north-west: the north side is nearest to you)



Compass: east is on the left of every picture and west on the right, so the sun rises on the left and sets on the right. Yellow floor = direct sun.

Sun and shadows through the day: 3D view from the north-west (front / street nearest).

The sun and its shadows through the day, seen from above (true north at the bottom)



Compass: east is on the left of every picture and west on the right, so the sun rises on the left and sets on the right. Yellow floor = direct sun.

The same moments seen from above (street / front at the bottom).

What counts as good? (light)

Daylight at the centre of the room (NBR 15575 test) Brazilian norm

Natural light on a table-height plane (0.75 m) at the middle of the room, on 23 April and 23 October at 9:30 and 15:30 with a half-cloudy sky, lamps off. The lowest of the four readings counts. Applies to living rooms, bedrooms, kitchens and service areas.

Fails 1/5	below 60 lux	Below the 60 lux minimum: you need a lamp even at midday.
Minimum 3/5	60 to 90 lux	Meets the legal minimum of the norm.
Intermediate 4/5	90 to 120 lux	Better than the minimum.
Superior 5/5	120 or more lux	Top level of the norm.

This house: Living (front half) 91 lux Intermediate 4/5 | TV / dining (rear half) 408 lux Superior 5/5 | Bedroom 1 513 lux Superior 5/5 | Bedroom 2 93 lux Intermediate 4/5 | Bedroom 3 123 lux Superior 5/5 | Kitchen 188 lux Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Our run uses the typical weather of the week around those dates and sky light only (no direct sun), so it is on the safe side.

Daylight at the centre of the room (NBR 15575 test) - bathrooms, hallways, stairs, garages Brazilian norm

Same test as above, but for rooms where people do not stay long.

No requirement 3/5	below 30 lux	The norm does not require daylight here (minimum level: 'not required').
Intermediate 4/5	30 to 45 lux	Meets the intermediate level.
Superior 5/5	45 or more lux	Meets the superior level.

This house: Hallway 0 lux No requirement 3/5 | Bathroom 1 42 lux Intermediate 4/5 | Bathroom 2 48 lux Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements.

Share of the floor that reaches the home target light level International standard

Percentage of the room's floor that gets at least the target light (100 lux bedroom, 150 lux living room, 200 lux kitchen) from daylight alone, for at least half of the daytime hours (8:00-18:00).

Below target 2/5	below 50 % of floor	Less than half of the room reaches the target: lamps will be needed in the daytime.
Meets target 4/5	50 to 95 % of floor	At least half of the floor reaches the target for half of the day.
Whole room 5/5	95 or more % of floor	Practically the entire room reaches the target.

This house: Living (front half) 26% Below target 2/5 | TV / dining (rear half) 100% Whole room 5/5 | Bedroom 1 100% Whole room 5/5 | Bedroom 2 100% Whole room 5/5 | Bedroom 3 100% Whole room 5/5 | Kitchen 100% Whole room 5/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks for the target over at least 50% of the area for half of the daylight hours (+ a lower level over 95%).

Spatial daylight autonomy sDA 300/50% (work-space standard) International standard

Percentage of the floor with at least 300 lux of daylight for at least half of the occupied hours. 300 lux is a WORKING-light level, so this is a strict check: useful for a home office, kitchen or study; bedrooms and hallways are not expected to reach it.

Below recommendation 2/5	below 55 % of floor	Less than 55% of the floor reaches 300 lux for half of the day.
Nominally acceptable 3/5	55 to 75 % of floor	Accepted by the IES / LEED (55% or more).
Preferred 5/5	75 or more % of floor	Preferred level (75% or more).

This house: Living (front half) 2% Below recommendation 2/5 | TV / dining (rear half) 100% Preferred 5/5 | Hallway 0% Below recommendation 2/5 | Bedroom 1 100% Preferred 5/5 | Bathroom 1 0% Below recommendation 2/5 | Bedroom 2 0% Below recommendation 2/5 | Bathroom 2 0% Below recommendation 2/5 | Bedroom 3 8% Below recommendation 2/5 | Kitchen 0% Below recommendation 2/5 | Outdoor bath 0% Below recommendation 2/5 | Outdoor storage 17% Below recommendation 2/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Annual sunlight exposure ASE (glare and overheating risk) International standard

Percentage of the floor that receives strong direct sun (over 1000 lux) for more than 250 hours a year. High values mean glare, faded furniture and heat.

Very low 5/5	below 3 % of floor	Practically no harsh direct sun.
Acceptable 4/5	3 to 10 % of floor	Within the 10% limit used by IES / LEED.
High 2/5	10 to 20 % of floor	Above the 10% limit: consider curtains, blinds or an eave.
Very high 1/5	20 or more % of floor	Large area with long strong sun: glare and heat likely.

This house: Living (front half) 0% Very low 5/5 | TV / dining (rear half) 0% Very low 5/5 | Hallway 0% Very low 5/5 | Bedroom 1 10% High 2/5 | Bathroom 1 0% Very low 5/5 | Bedroom 2 0% Very low 5/5 | Bathroom 2 0% Very low 5/5 | Bedroom 3 0% Very low 5/5 | Kitchen 0% Very low 5/5 | Outdoor bath 0% Very low 5/5 | Outdoor storage 0% Very low 5/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Daytime hours without needing lamps Rule of thumb

Share of the daytime hours in which the room's average daylight already reaches the recommended general light level (see the lamp table), so the lights can stay off.

Almost never 1/5	below 10 % of 8:00-18:00 hours	You will switch the lights on all day.
Rarely 2/5	10 to 25 % of 8:00-18:00 hours	Lamps needed most of the day.
Sometimes 3/5	25 to 50 % of 8:00-18:00 hours	Lamps needed on dull days and mornings / evenings.
Often 4/5	50 to 75 % of 8:00-18:00 hours	Lamps mostly off during the day.
Almost always 5/5	75 or more % of 8:00-18:00 hours	Lamps off nearly the whole day.

This house: Living (front half) 47% Sometimes 3/5 | TV / dining (rear half) 96% Almost always 5/5 | Hallway 0% Almost never 1/5 | Bedroom 1 96% Almost always 5/5 | Bathroom 1 1% Almost never 1/5 | Bedroom 2 40% Sometimes 3/5 | Bathroom 2 0% Almost never 1/5 | Bedroom 3 70% Often 4/5 | Kitchen 81% Almost always 5/5 | Outdoor bath 14% Rarely 2/5 | Outdoor storage 78% Almost always 5/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels).

Direct sun on the window in early spring International standard

Hours of direct sun that reach the room's windows on a typical day in early spring: 1 August to 21 September in the southern hemisphere, 1 February to 21 March in the northern one (the EN 17037 dates). Sun in winter / early spring is welcome for health and warmth.

Below minimum 2/5	below 1.5 hours/day	Less than 1.5 h of sun: the room feels sunless.
Minimum 3/5	1.5 to 3 hours/day	Meets the EN 17037 minimum (1.5 h).
Medium 4/5	3 to 4 hours/day	Meets the EN 17037 medium level (3 h).
High 5/5	4 or more hours/day	Meets the EN 17037 high level (4 h).

This house: Living (front half) 0.0 h Below minimum 2/5 | TV / dining (rear half) 0.3 h Below minimum 2/5 | Bedroom 1 3.7 h Medium 4/5 | Bedroom 2 0.0 h Below minimum 2/5 | Bedroom 3 0.0 h Below minimum 2/5 | Kitchen 0.0 h Below minimum 2/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks this for at least one main room (preferably the living room); bathrooms and hallways are exempt.

3. Lamps: how much light to install (lumens)

Lux is the light that reaches a surface; **lumens** is the light a lamp gives out. To get an average of X lux over a floor of A m² you need X × A ÷ (utilisation × maintenance = 0.45 × 0.8 = 0.36) lumens. The 'minimum' column is the bare lux × m².

3.1 Recommended light levels at home

Kind of room	General light (lux): min / recommended / max	Task light (lux)	Source
Living / dining room	100 / 150 / 200	300 / 500 / 750 - reading, writing, sewing	NBR 5413 (home); EN 12464-1 / NBR 8995-1 for reading tasks
Bedroom	100 / 150 / 200	200 / 300 / 500 - reading in bed, dressing table	NBR 5413 (home)
Kitchen	100 / 150 / 200	200 / 300 / 500 - stove, sink and worktop (EN 12464-1 asks 500)	NBR 5413 (home); EN 12464-1
Bathroom	100 / 150 / 200	200 / 300 / 500 - mirror (shaving, make-up)	NBR 5413 (home)
Hallway / circulation	75 / 100 / 150	-	NBR 5413 (home)
Laundry / storage / outdoor bath	100 / 150 / 200	200 / 300 / 500 - laundry and tools	NBR 5413 (home)
Porch / carport (covered outdoor)	50 / 75 / 100	-	NBR 5413 (garages 50-75-100)

Calculated with: **3D model (IFC) and tables** · norm tables, no calculation · details and reliability in section 9

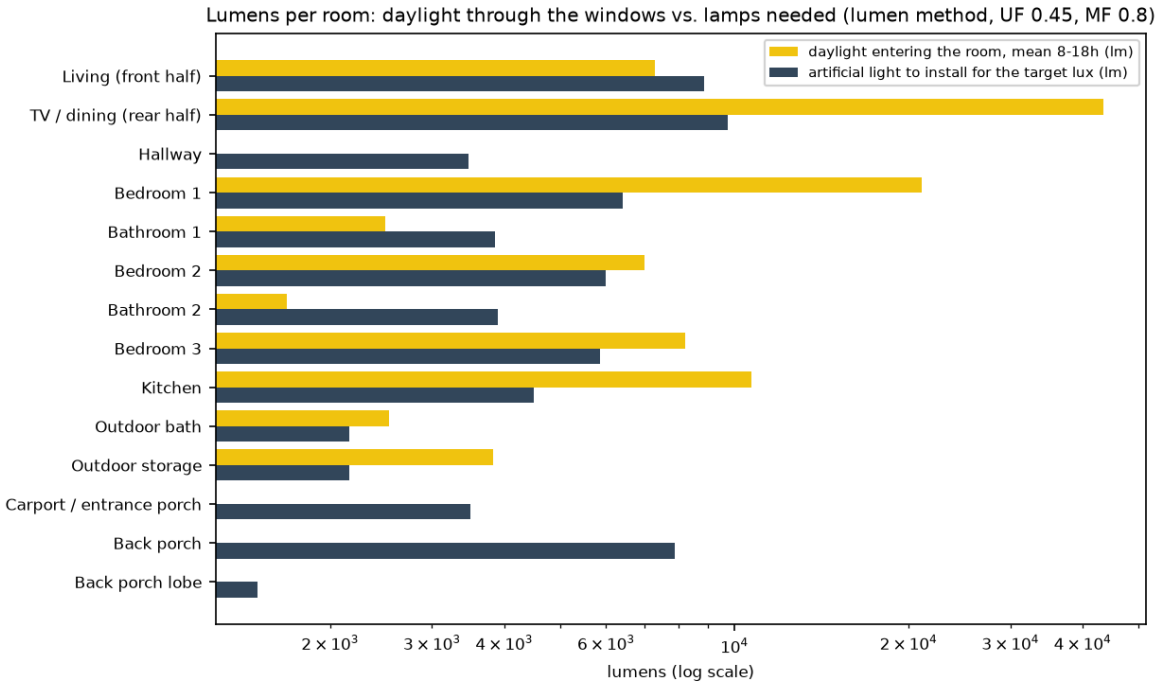
Reading: light the room to the general level (100-200 lux) and add a directional **reading lamp of about 450-800 lumens** (about 4-8 W LED) near the page to reach **300-500 lux** on it (NBR ISO/CIE 8995-1 and EN 12464-1 for reading and writing; NBR 5413 for homes). Reading under only the 100-150 lux of general light strains the eyes. For scale: 100 lm = night light / small candle-type; 450 lm = equivalent of an old 40 W bulb; 800 lm = equivalent of an old 60 W bulb; 1600 lm = equivalent of an old 100 W bulb.

3.2 Lamp lumens for each room of this house

Room	m²	General light goal	Task light (reading etc.)	Lamp lumens: minimum - design	LED watts (design)
Living (front half)	21.2	150 lux	500 lux (reading, writing, sewing)	3,186 - 8,850	88
TV / dining (rear half)	23.4	150 lux	500 lux (reading, writing, sewing)	3,510 - 9,750	98
Hallway	12.5	100 lux	-	1,248 - 3,467	35
Bedroom 1	15.4	150 lux	300 lux (reading in bed, dressing table)	2,310 - 6,417	64
Bathroom 1	9.2	150 lux	300 lux (mirror (shaving, make-up))	1,386 - 3,850	38
Bedroom 2	14.4	150 lux	300 lux (reading in bed, dressing table)	2,160 - 6,000	60
Bathroom 2	9.4	150 lux	300 lux (mirror (shaving, make-up))	1,404 - 3,900	39
Bedroom 3	14.1	150 lux	300 lux (reading in bed, dressing table)	2,109 - 5,858	59
Kitchen	10.8	150 lux	300 lux (stove, sink and worktop (EN 12464-1 asks 500))	1,620 - 4,500	45
Outdoor bath	5.2	150 lux	300 lux (laundry and tools)	778 - 2,160	22
Outdoor storage	5.2	150 lux	300 lux (laundry and tools)	778 - 2,160	22

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

Use the low end for light walls and ceilings and good fittings; the high end for dark colours or recessed spots. LED at 100 lm per watt. Daylight already covers part of the day (section 2.2).



Daylight entering through the windows (yellow) compared with the lamp lumens needed for the target light (dark).

4. Windows, fresh air and wind

NBR 15575-4 asks that the openable part of the windows is at least **7% of the floor area** of each living room and bedroom, and limits glass to 20% of the floor unless it is shaded. NBR 15220-3 recommends wind-driven cross-ventilation as the main summer strategy for this climate.

4.1 Window size

Room	Floor m²	Glass m²	Glass / floor	Openable / floor
Living + dining	44.6	6.64	14.9% Small 3/5	7.4% Meets minimum 3/5
Hallway	12.5	0.00	0.0%	0.0%
Bedroom 1	15.4	1.44	9.4% Small 3/5	4.7% Fails 1/5
Bathroom 1	9.2	0.36	3.9%	1.9%
Bedroom 2	14.4	1.44	10.0% Small 3/5	5.0% Fails 1/5
Bathroom 2	9.4	0.36	3.8%	1.9%
Bedroom 3	14.1	1.44	10.2% Small 3/5	5.1% Fails 1/5
Kitchen	10.8	1.80	16.7% Ideal 5/5	8.3% Meets minimum 3/5
Outdoor bath	5.2	0.36	6.9%	3.5%
Outdoor storage	5.2	0.36	6.9%	3.5%

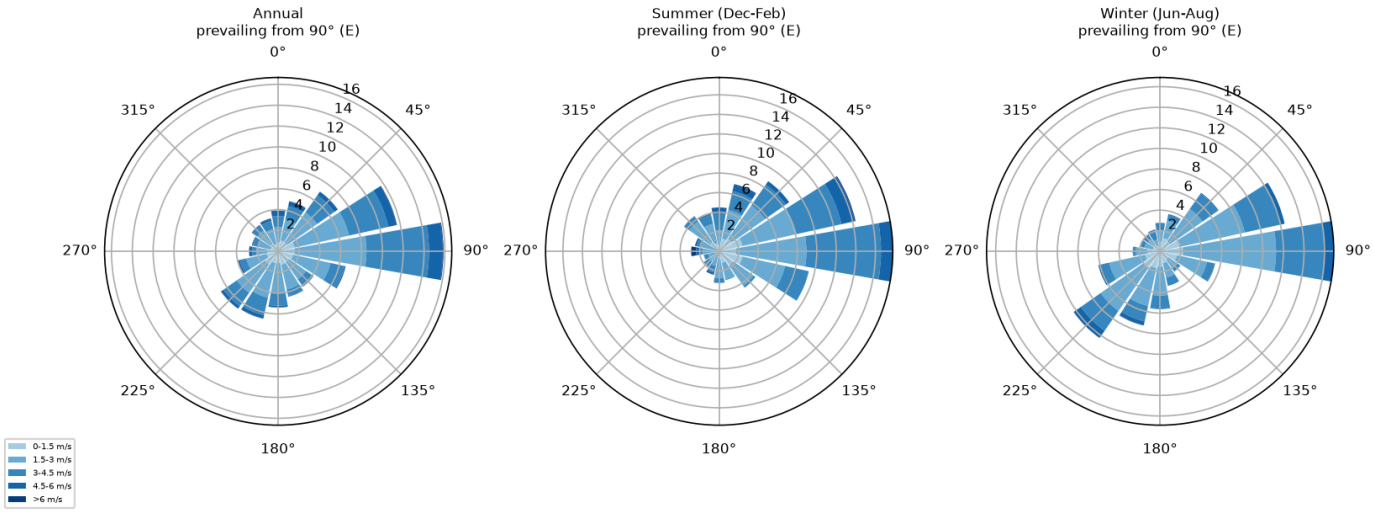
Calculated with: **3D model (IFC) and tables** · sizes read from the IFC model · details and reliability in section 9

Window sizes in the model are common catalogue sizes of about 2012 (bedrooms 1.20 x 1.20 m, living and dining 2.00 x 1.20 m, kitchen 1.50 x 1.20 m, bathrooms 0.60 x 0.60 m), not measurements; a sliding window is assumed to open half of its area. If the real windows differ, change them and rerun the analysis.

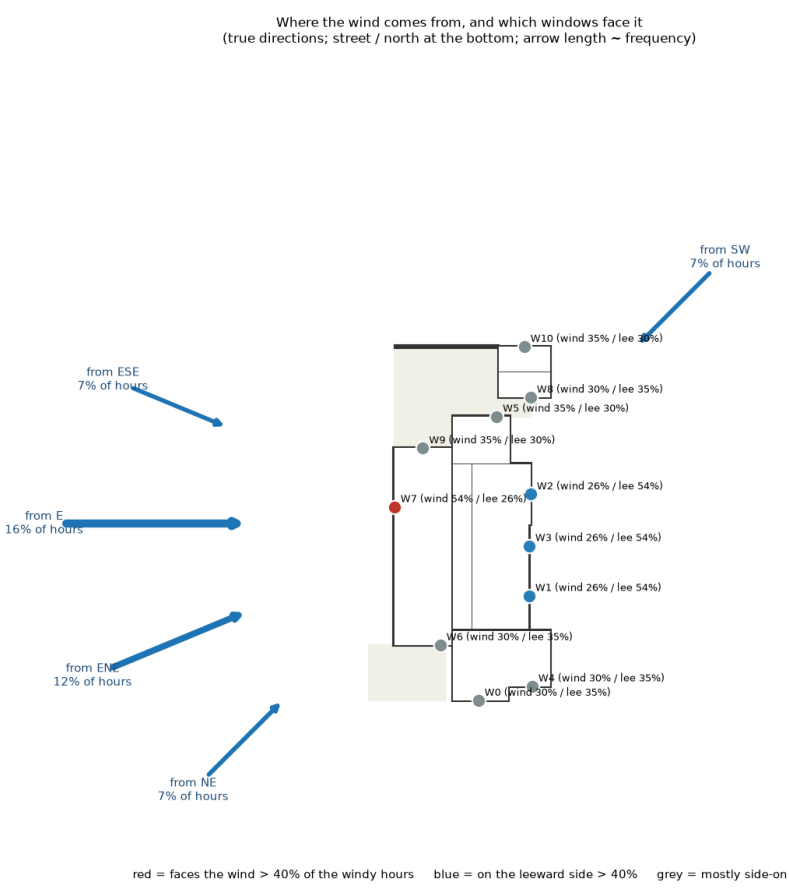
4.2 Cross-ventilation

Room	Windy hours with a through-draught
Living + dining	100% Excellent 5/5
Hallway	100%
Bedroom 1	100% Excellent 5/5
Bathroom 1	0%
Bedroom 2	100% Excellent 5/5
Bathroom 2	0%
Bedroom 3	100% Excellent 5/5
Kitchen	100% Excellent 5/5
Outdoor bath	0%
Outdoor storage	0%

Calculated with: **Fast engine (Python, this project)** · wind-hours rule from the weather file; no pressure calculation (the pressure-based ventilation is in section 6.4) · details and reliability in section 9



Wind roses: where the wind comes from and how strong, whole year, summer and winter.



Wind directions against the windows of the house (red windows face the wind, blue are on the sheltered side).

What counts as good? (windows and wind)

Openable window area for ventilation Brazilian norm

Area of windows that can really open (glass, frames and fixed leaves are discounted), divided by the floor area of the room. Zones 1-7 need at least 7%.

Fails 1/5	below 7 % of floor area	Less than the 7% minimum of the norm (unless the city's building code says otherwise).
Meets minimum 3/5	7 to 10 % of floor area	Meets the 7% minimum.
Comfortable margin 5/5	10 or more % of floor area	More than the minimum: good air changes.

This house: Living + dining 7.4% Meets minimum 3/5 | Bedroom 1 4.7% Fails 1/5 | Bedroom 2 5.0% Fails 1/5 | Bedroom 3 5.1% Fails 1/5 | Kitchen 8.3% Meets minimum 3/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas. We assume a sliding window opens about half of its area; a casement / tilting window opens more.

Glass area compared with the floor Brazilian norm

Window area divided by floor area. NBR 15575-4 limits transparent areas to 20% of the floor (rooms up to 20 m2) unless there is shading or high-performance glass; NBR 15220-3 recommends 15-25% openings for zone 3.

Very small 2/5	below 7 % of floor area	Too little window for light and air.
Small 3/5	7 to 15 % of floor area	Works, but a little dark or stuffy.
Ideal 5/5	15 to 20 % of floor area	Inside the NBR 15575-4 limit and the NBR 15220-3 recommendation.
Large 4/5	20 to 25 % of floor area	Recommended by NBR 15220-3 if shaded; above the NBR 15575-4 20% unless shaded.
Too large 2/5	25 or more % of floor area	More sun heat than the norms accept without shading.

This house: Living + dining 14.9% Small 3/5 | Bedroom 1 9.4% Small 3/5 | Bedroom 2 10.0% Small 3/5 | Bedroom 3 10.2% Small 3/5 | Kitchen 16.7% Ideal 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas; ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3).

Cross-ventilation possible Rule of thumb

Share of the windy hours in which the wind hits a window on one side of the connected rooms and another window on the opposite side can let it out (doors assumed open).

Almost none 1/5	below 5 % of windy hours	Air hardly flows through the room.
Weak 2/5	5 to 20 % of windy hours	Cross-ventilation seldom happens.
Some 3/5	20 to 40 % of windy hours	Works in part of the windy hours.
Good 4/5	40 to 60 % of windy hours	Works in many windy hours.
Excellent 5/5	60 or more % of windy hours	A through-draught is possible most of the time.

This house: Living + dining 100% Excellent 5/5 | Bedroom 1 100% Excellent 5/5 | Bedroom 2 100% Excellent 5/5 | Bedroom 3 100% Excellent 5/5 | Kitchen 100% Excellent 5/5

Source: Rule of thumb of this study (no norm gives a number).

Average outdoor wind speed International standard

Average wind at the site over the year (weather file, open terrain at 10 m; at the house it is lower).

Calm 2/5	below 1.0 m/s	Little natural ventilation from the wind alone.
Light air 3/5	1.0 to 2.0 m/s	Gentle; openings help but cross-ventilation is weak.
Light to gentle breeze 5/5	2.0 to 5.0 m/s	Ideal range for natural ventilation.
Moderate breeze 4/5	5.0 to 8.0 m/s	Strong enough to need wind protection outdoors.
Fresh breeze or more 2/5	8.0 or more m/s	Wind damage and discomfort risk.

This house: Maringá, PR, whole year 2.3 m/s Light to gentle breeze 5/5

Source: WMO Beaufort wind scale; Rule of thumb of this study (no norm gives a number). Beaufort 1 = 0.3-1.5 m/s, 2 = 1.6-3.3, 3 = 3.4-5.4, 4 = 5.5-7.9 m/s. Indoors ASHRAE-55 accepts air speeds up to about 0.8 m/s.

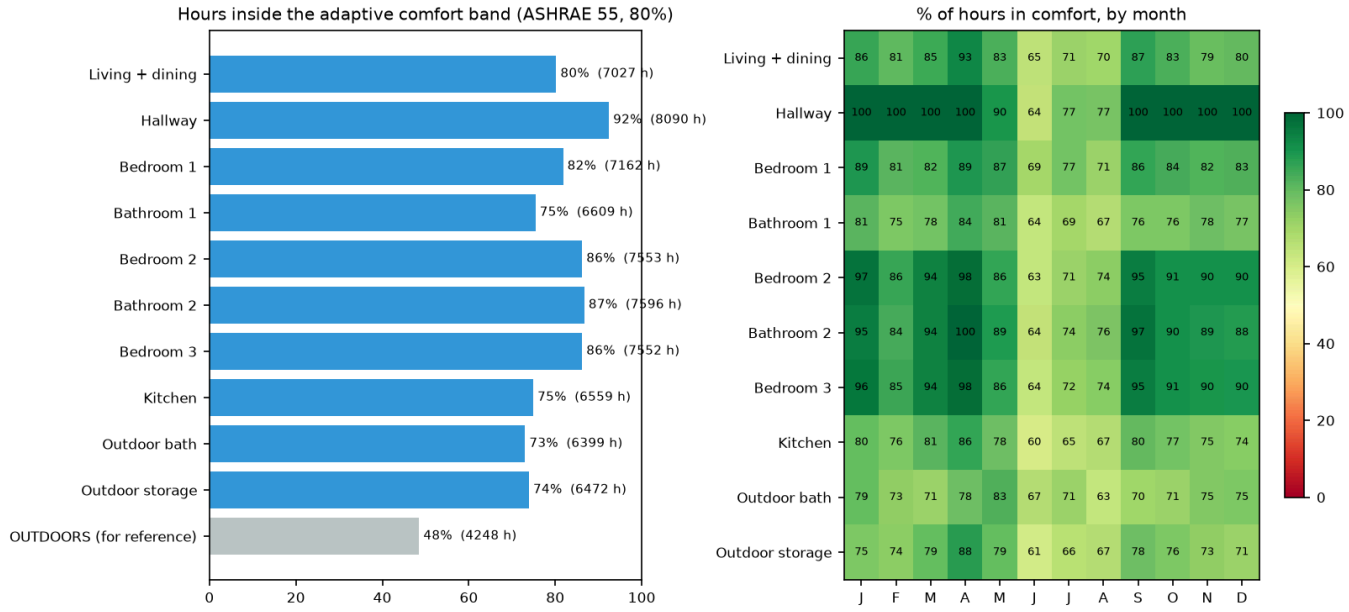
5. Temperature and comfort

Without air conditioning, comfort depends on the walls, roof, windows, shade and ventilation. The simulation computes the indoor temperature of every room for each of the 8,760 hours of a typical year. **Comfortable** means inside the ASHRAE-55 adaptive comfort band (it moves with the weather: people accept warmer rooms when it is warmer outside).

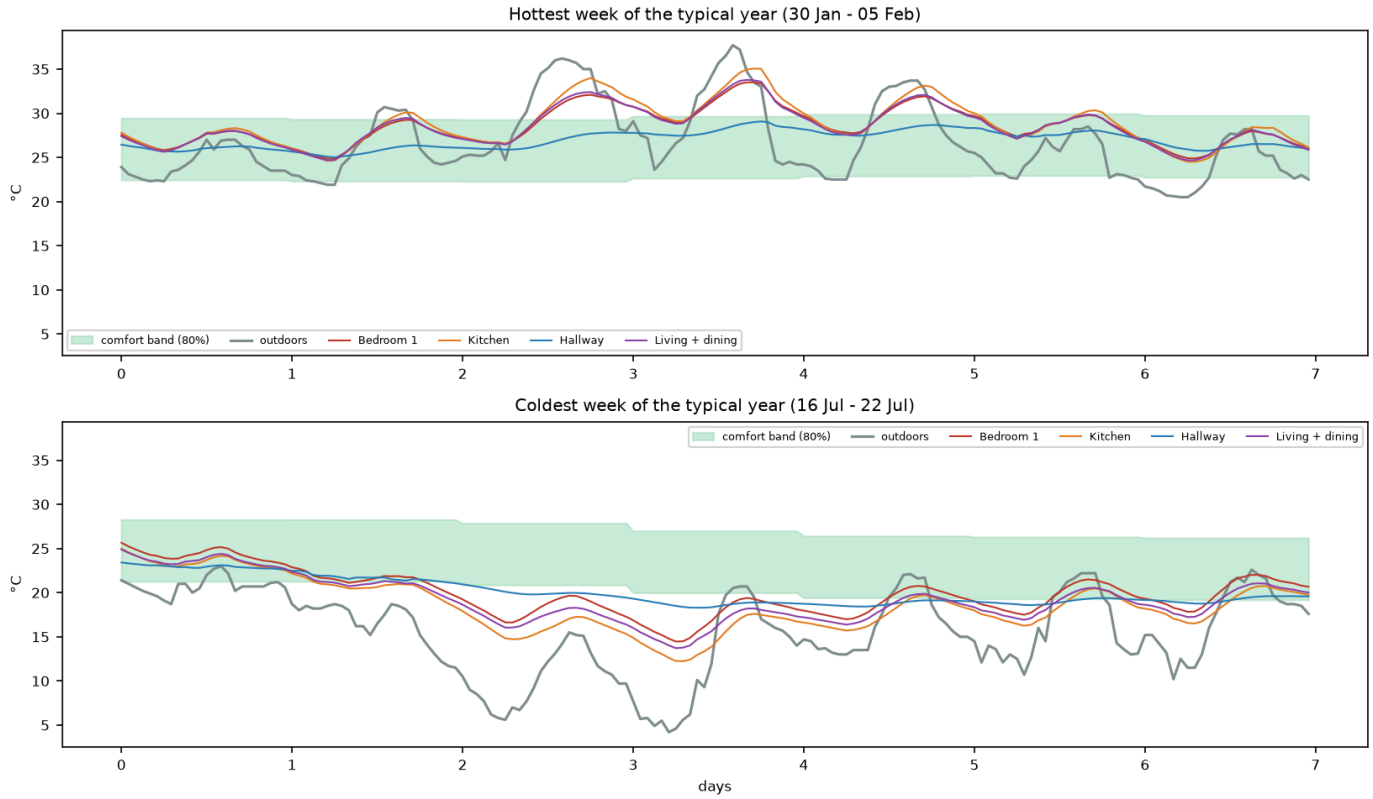
5.1 Hours in the comfortable band

Room	Comfortable hours	Too hot	Too cold	Hottest	Coldest
Living + dining	7,027 h = 80% Good 4/5	842 h	891 h	33.8 °C Very hot 2/5	12.8 °C Cold 2/5
Hallway	8,090 h = 92% Excellent 5/5	0 h	670 h	29.1 °C Warm 4/5	16.5 °C Cool 3/5
Bedroom 1	7,162 h = 82% Good 4/5	875 h	723 h	33.5 °C Very hot 2/5	13.0 °C Cold 2/5
Bathroom 1	6,609 h = 75% Good 4/5	1,223 h	928 h	34.9 °C Very hot 2/5	11.7 °C Dangerous 1/5
Bedroom 2	7,553 h = 86% Good 4/5	402 h	805 h	32.8 °C Very hot 2/5	14.0 °C Cold 2/5
Bathroom 2	7,596 h = 87% Good 4/5	450 h	714 h	32.5 °C Very hot 2/5	14.4 °C Cold 2/5
Bedroom 3	7,552 h = 86% Good 4/5	409 h	799 h	32.8 °C Very hot 2/5	14.0 °C Cold 2/5
Kitchen	6,559 h = 75% Acceptable 3/5	1,109 h	1,092 h	35.1 °C Extreme 1/5	11.5 °C Dangerous 1/5
Outdoor bath	6,399 h = 73% Acceptable 3/5	1,546 h	815 h	35.1 °C Extreme 1/5	11.5 °C Dangerous 1/5
Outdoor storage	6,472 h = 74% Acceptable 3/5	1,258 h	1,030 h	35.1 °C Extreme 1/5	11.3 °C Dangerous 1/5
OUTDOORS (for reference)	4,248 h = 48% Poor 2/5	1,193 h	3,319 h	37.7 °C	4.2 °C

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9



Left: comfortable hours per room. Right: share of comfortable hours by month (green = good, red = bad).



A hot week and a cold week: indoor temperatures against the comfort band and the outdoor temperature.

5.2 Brazilian design-day check (NBR 15575)

Hottest day of the typical year: **02 Feb** (37.7 °C outside). Coldest day: **19 Jul** (4.2 °C outside).

Room	Hottest day: indoor peak	vs. outdoors	Verdict	Coldest day: indoor minimum	vs. outdoors	Verdict
Living + dining	33.9 °C	-3.8	Intermediate 4/5	13.5 °C	+9.3	Superior 5/5
Hallway *(not rated)*	29.0 °C	-8.7		18.3 °C	+14.1	
Bedroom 1	33.6 °C	-4.1	Superior 5/5	14.3 °C	+10.1	Superior 5/5
Bathroom 1 *(not rated)*	35.0 °C	-2.7		12.4 °C	+8.2	
Bedroom 2	32.8 °C	-4.9	Superior 5/5	14.9 °C	+10.7	Superior 5/5
Bathroom 2 *(not rated)*	32.5 °C	-5.2		15.6 °C	+11.4	
Bedroom 3	32.8 °C	-4.9	Superior 5/5	15.0 °C	+10.8	Superior 5/5
Kitchen *(not rated)*	35.1 °C	-2.6		12.0 °C	+7.8	
Outdoor bath *(not rated)*	35.1 °C	-2.6		12.6 °C	+8.4	
Outdoor storage *(not rated)*	35.2 °C	-2.5		11.9 °C	+7.7	

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

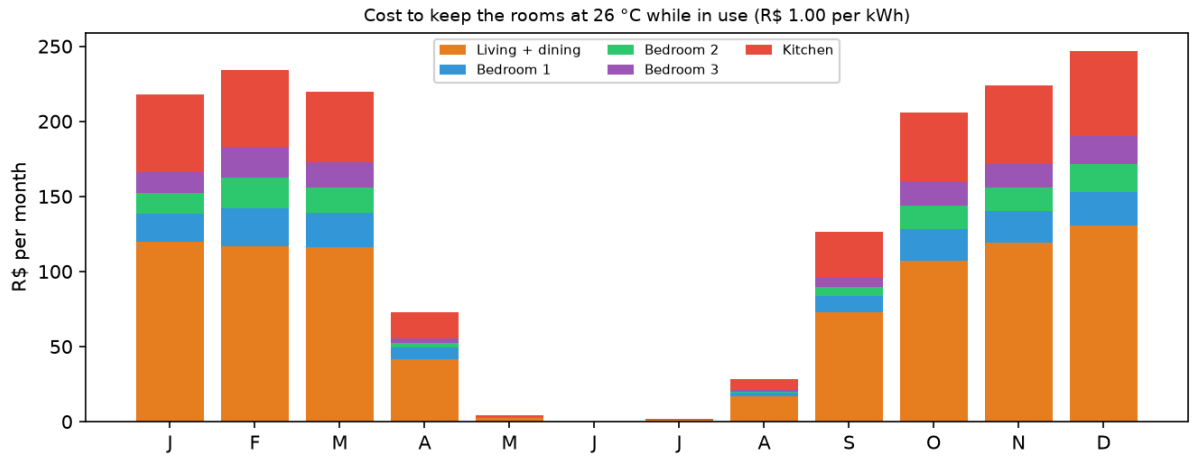
The newest edition (NBR 15575-1:2021) compares the house with a reference house over the whole year; that needs a second simulation, so this report uses the classic absolute design-day criteria, which are easier to understand.

5.3 What it costs to cool the rooms in summer

If a room is too hot you can cool it with an air conditioner. The simulation switches on a split air conditioner whenever the room is above **26 °C** during the hours it is used (bedroom: 21:00-08:00, living: 14:00-23:00, kitchen: 08:00-21:00), with windows closed, and adds up the electricity: heat removed ÷ COP 3.0 (an inverter model does better) × tariff **R$ 1.00 per kWh** (all taxes in; an assumption based on the Copel residential tariff after the +20% adjustment of June 2026; change it with *run_all.bat --tariff 0.95*).

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

Room	When it is cooled	Hours the A/C runs	Electricity kWh/year	Cost R$/year	Cost Dec-Feb	Worst month	Peak heat to remove	A/C size	Level
Living + dining (44.6 m²)	14:00-23:00	1627	844	**R$ 844**	R$ 367	Dec (R$ 131)	5.8 kW	24,000 BTU/h	Low 4/5
Bedroom 1 (15.4 m²)	21:00-08:00	975	153	**R$ 153**	R$ 66	Feb (R$ 25)	1.9 kW	9,000 BTU/h	Very low 5/5
Bedroom 2 (14.4 m²)	21:00-08:00	875	112	**R$ 112**	R$ 53	Feb (R$ 21)	1.6 kW	9,000 BTU/h	Very low 5/5
Bedroom 3 (14.1 m²)	21:00-08:00	887	110	**R$ 110**	R$ 53	Feb (R$ 20)	1.6 kW	9,000 BTU/h	Very low 5/5
Kitchen (10.8 m²)	08:00-21:00	2157	363	**R$ 363**	R$ 160	Dec (R$ 57)	1.5 kW	9,000 BTU/h	Moderate 3/5
Total			**1,582**	**R$ 1,582**					



Monthly cost of cooling by room.

'Peak heat to remove' is the cooling power needed in the hottest hour; the A/C size adds 10% and rounds up to a catalogue size (1 kW = 3,412 BTU/h). Only sensible heat is counted (no dehumidification), so real bills are a little higher. Rooms with a High or Very high level are the ones where shade, a better roof or better insulation pay back fastest.

What counts as good? (temperature, cooling and heating)

Hours in the comfortable temperature band Rule of thumb

Share of the 8,760 hours of the year in which the room temperature stays inside the ASHRAE-55 adaptive comfort band (it moves with the weather: warmer outdoors = wider acceptance). Outdoors, for reference, only about half of the hours.

Bad 1/5	below 40 % of the year	Uncomfortable most of the year.
Poor 2/5	40 to 60 % of the year	Uncomfortable much of the year.
Acceptable 3/5	60 to 75 % of the year	Comfortable most of the time, with some hot or cold spells.
Good 4/5	75 to 90 % of the year	Comfortable most of the year.
Excellent 5/5	90 or more % of the year	Comfortable nearly all year without air conditioning or heating.

This house: Living + dining 80% Good 4/5 | Hallway 92% Excellent 5/5 | Bedroom 1 82% Good 4/5 | Bathroom 1 75% Good 4/5 | Bedroom 2 86% Good 4/5 | Bathroom 2 87% Good 4/5 | Bedroom 3 86% Good 4/5 | Kitchen 75% Acceptable 3/5 | Outdoor bath 73% Acceptable 3/5 | Outdoor storage 74% Acceptable 3/5 | OUTDOORS (for reference) 48% Poor 2/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The norms judge thermal comfort relative to a reference house or on design days; this percentage scale is our own.

Hottest day: indoor peak vs. outdoor peak (NBR 15575) Brazilian norm

On the hottest day of the typical year, the hottest indoor air temperature compared with the hottest outdoor temperature. Negative means the house is cooler than the outdoors.

Superior 5/5	below -4 degC (indoor max - outdoor max)	Indoor peak at least 4 degC below the outdoor peak.
Intermediate 4/5	-4 to -2 degC (indoor max - outdoor max)	Indoor peak at least 2 degC below the outdoor peak.
Minimum 3/5	-2 to 0.0001 degC (indoor max - outdoor max)	Indoor peak no higher than the outdoor peak.
Fails 1/5	0.0001 or more degC (indoor max - outdoor max)	Indoor peak higher than outdoors: the house overheats.

This house: Living + dining -3.8 °C Intermediate 4/5 | Bedroom 1 -4.1 °C Superior 5/5 | Bedroom 2 -4.9 °C Superior 5/5 | Bedroom 3 -4.9 °C Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Applies to bedrooms and living rooms. We use the hottest day of the weather file, not the norm's standard design day.

Coldest day: indoor minimum vs. outdoor minimum (NBR 15575) Brazilian norm

On the coldest day of the typical year, the coldest indoor air temperature compared with the coldest outdoor temperature. Positive means the house is warmer than the outdoors.

Fails 1/5	below 3 degC (indoor min - outdoor min)	Less than 3 degC warmer than outdoors.
Minimum 3/5	3 to 5 degC (indoor min - outdoor min)	At least 3 degC warmer than outdoors.
Intermediate 4/5	5 to 7 degC (indoor min - outdoor min)	At least 5 degC warmer than outdoors.
Superior 5/5	7 or more degC (indoor min - outdoor min)	At least 7 degC warmer than outdoors.

This house: Living + dining +9.3 °C Superior 5/5 | Bedroom 1 +10.1 °C Superior 5/5 | Bedroom 2 +10.7 °C Superior 5/5 | Bedroom 3 +10.8 °C Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Winter check is required in bioclimatic zones 1 to 5, which includes Maringa.

Hottest hour of the year inside Rule of thumb

Highest operative temperature (air + wall surfaces) reached in the room.

Comfortable 5/5	below 28 degC	Never really hot.
Warm 4/5	28 to 30 degC	Warm on the hottest days; a fan is enough.
Hot 3/5	30 to 32 degC	Hot spells; fans help, sleep is affected.
Very hot 2/5	32 to 35 degC	Heat stress risk for children and the elderly.
Extreme 1/5	35 or more degC	Dangerous without cooling.

This house: Living + dining 33.8 °C Very hot 2/5 | Hallway 29.1 °C Warm 4/5 | Bedroom 1 33.5 °C Very hot 2/5 | Bathroom 1 34.9 °C Very hot 2/5 | Bedroom 2 32.8 °C Very hot 2/5 | Bathroom 2 32.5 °C Very hot 2/5 | Bedroom 3 32.8 °C Very hot 2/5 | Kitchen 35.1 °C Extreme 1/5 | Outdoor bath 35.1 °C Extreme 1/5 | Outdoor storage 35.1 °C Extreme 1/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The adaptive-comfort upper limit in a warm month is about 29 degC (80% acceptability).

Coldest hour of the year inside International standard

Lowest operative temperature reached in the room.

Dangerous 1/5	below 12 degC	Cold stress; risk of heart and breathing problems.
Cold 2/5	12 to 16 degC	Below 16 degC the risk of respiratory problems rises.
Cool 3/5	16 to 18 degC	A jumper is needed; below the WHO 18 degC recommendation.
Good 4/5	18 to 20 degC	Meets the WHO recommendation of at least 18 degC.
Pleasant 5/5	20 or more degC	Never uncomfortably cold.

This house: Living + dining 12.8 °C Cold 2/5 | Hallway 16.5 °C Cool 3/5 | Bedroom 1 13.0 °C Cold 2/5 | Bathroom 1 11.7 °C Dangerous 1/5 | Bedroom 2 14.0 °C Cold 2/5 | Bathroom 2 14.4 °C Cold 2/5 | Bedroom 3 14.0 °C Cold 2/5 | Kitchen 11.5 °C Dangerous 1/5 | Outdoor bath 11.5 °C Dangerous 1/5 | Outdoor storage 11.3 °C Dangerous 1/5

Source: WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C).

Electricity to keep the room cool with an air conditioner Rule of thumb

Electricity a split air conditioner would use in a year to hold the room at 26 C (upper limit of the summer comfort zone of NBR 16401-2) during the hours the room is used (bedrooms 21:00-08:00, living 14:00-23:00, kitchen 08:00-21:00). A room that overheats in summer costs more here. Cost = kWh x tariff.

Very low 5/5	below 10 kWh per m2 per year	Rarely needs cooling.
Low 4/5	10 to 25 kWh per m2 per year	A few weeks of cooling a year.
Moderate 3/5	25 to 45 kWh per m2 per year	Cooling needed through most of the summer.
High 2/5	45 to 70 kWh per m2 per year	Expensive to keep cool: the room overheats a lot.
Very high 1/5	70 or more kWh per m2 per year	Very expensive: shade, insulation or a better roof are worth more than a bigger air conditioner.

This house: Living + dining 18.9 kWh/m^2 Low 4/5 | Bedroom 1 9.9 kWh/m^2 Very low 5/5 | Bedroom 2 7.7 kWh/m^2 Very low 5/5 | Bedroom 3 7.8 kWh/m^2 Very low 5/5 | Kitchen 33.6 kWh/m^2 Moderate 3/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH. Assumes COP 3.0 (Inmetro class A split of about 2012), windows closed while the air conditioner runs, sensible heat only (no dehumidification) so real bills are somewhat higher. Tariff is an assumption, see the report.

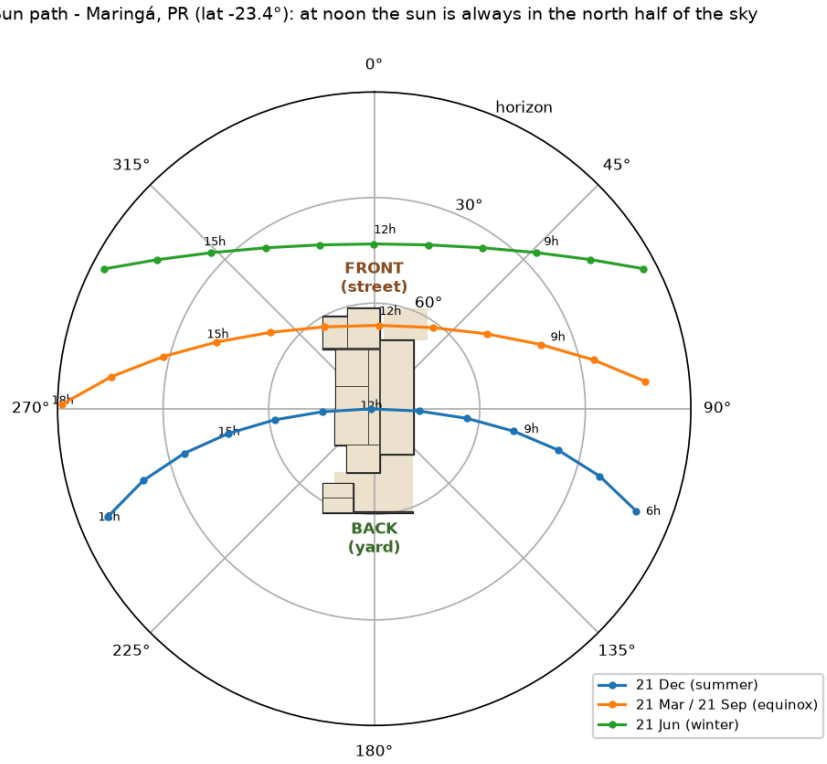
6. Sun on the outside, walls and roof

At this latitude the midday sun is always in the north half of the sky, so the walls that face north get the winter sun (good) and little summer sun, while east and west walls take strong summer sun (bad).

6.1 Sun energy on each surface (kWh per m², no shading)

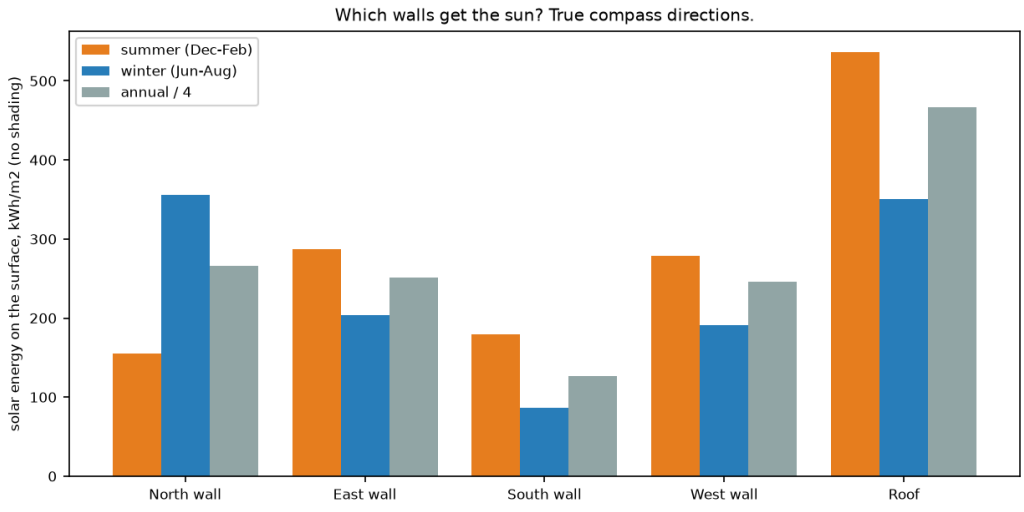
Surface	Whole year	Summer (Dec-Feb)	Winter (Jun-Aug)
North wall	1066	155 Moderate 4/5	356 High 5/5
East wall	1004	287 High 3/5	204 Moderate 3/5
South wall	507	179 Moderate 4/5	86 Low 2/5
West wall	982	279 High 3/5	191 Moderate 3/5
Roof (horizontal)	1865	536	351

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9



The house plan is drawn at the centre in its real orientation; its size is not to scale with the sky angles.

Path of the sun over the year with the house drawn at the centre in its real orientation: the front (street) faces north, the back faces south.



Solar energy on each wall by season.

6.2 Do the walls and roof meet NBR 15575?

Item	This house	Limit (NBR 15575)	Result
External wall heat transmittance U Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	2.52 W/m²K	<= 3.7	meets
External wall heat capacity Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	134 kJ/m²K	>= 130	meets
Roof heat transmittance U (indicative) Clay tile + ventilated attic + 12 cm concrete slab ceiling (paint / roof absorptance 0.7)	1.78 W/m²K	<= 1.5	does not meet

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

6.3 What if the materials change?

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

An existing house rarely gets new walls, but the roof / ceiling can often be improved, so the two are shown separately (each with the other part kept as it is today). Roof upgrades such as wool insulation are in section 7.

Walls

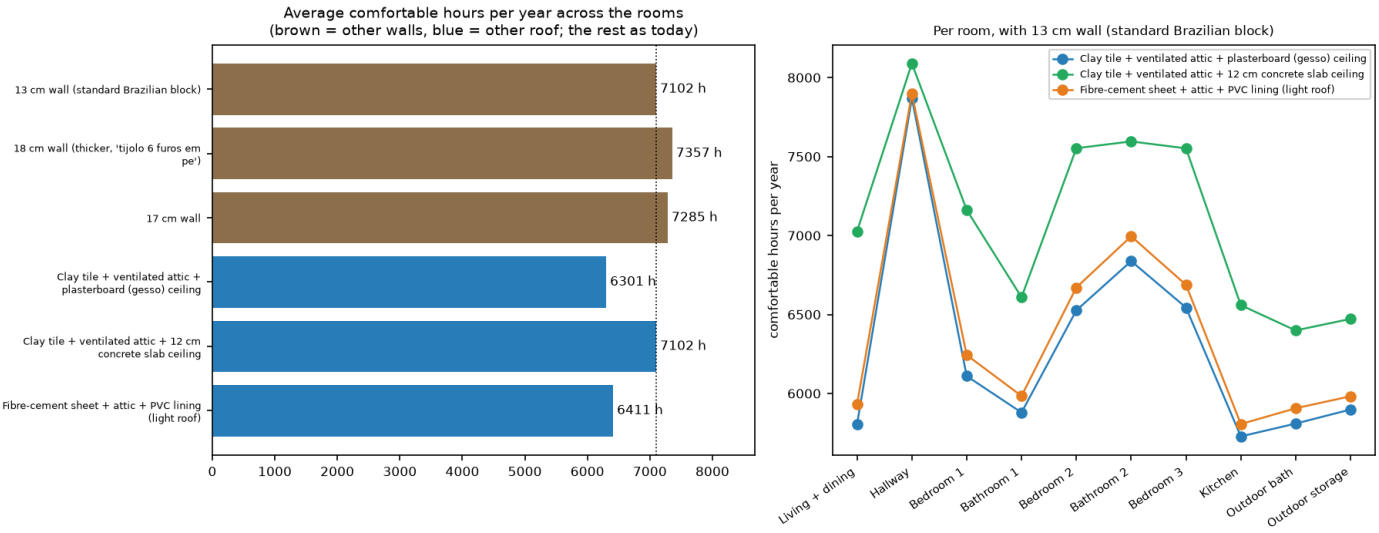
Roof / ceiling kept: Clay tile + ventilated attic + 12 cm concrete slab ceiling

Walls	Avg. comfortable hours	Too hot	Too cold	Cooling bill	vs. current
Ceramic hollow brick 14 cm + 2 cm plaster both sides = 18 cm wall (thicker, 'tijolo 6 furos em pe')	7,357 h = 84% Good 4/5	627 h	776 h	R$ 1,358/yr	R$ -224/yr
Concrete block 14 cm + 1.5 cm plaster both sides = 17 cm wall	7,285 h = 83% Good 4/5	694 h	781 h	R$ 1,543/yr	R$ -39/yr
Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) **(current)**	7,102 h = 81% Good 4/5	811 h	847 h	R$ 1,582/yr	-

Roof / ceiling

Walls kept: Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)

Roof / ceiling	Avg. comfortable hours	Too hot	Too cold	Cooling bill	vs. current
Clay tile + ventilated attic + 12 cm concrete slab ceiling **(current)**	7,102 h = 81% Good 4/5	811 h	847 h	R$ 1,582/yr	-
Fibre-cement sheet + attic + PVC lining (light roof)	6,411 h = 73% Acceptable 3/5	1,180 h	1,169 h	R$ 1,798/yr	+R$ 216/yr
Clay tile + ventilated attic + plasterboard (gesso) ceiling	6,301 h = 72% Acceptable 3/5	1,304 h	1,156 h	R$ 1,958/yr	+R$ 376/yr



Comfortable hours with different walls and roofs.

What counts as good? (sun and construction)

Summer sun on a wall Rule of thumb

Solar energy on a vertical wall with no shading in the three summer months (Dec-Feb in the south, Jun-Aug in the north). Lower is better: it is heat you do not want.

Low 5/5	below 150 kWh/m2	Little summer heat from the sun.
Moderate 4/5	150 to 225 kWh/m2	Some heat; an eave or a light colour handles it.
High 3/5	225 to 300 kWh/m2	Hot wall: needs shading, light colour, or insulation.
Very high 2/5	300 or more kWh/m2	Very hot wall.

This house: North wall 155 Moderate 4/5 | East wall 287 High 3/5 | South wall 179 Moderate 4/5 | West wall 279 High 3/5

Source: Rule of thumb of this study (no norm gives a number). Roughly 1.7 / 2.5 / 3.3 kWh/m2 per day. The norm handles wall heat through limits on U-value and paint colour.

Winter sun on a wall Rule of thumb

Solar energy on a vertical wall in June-August. Higher is better in a cool winter: free heat (NBR 15220-3 recommends winter solar heating for zone 3).

Low 2/5	below 120 kWh/m2	Almost no winter sun.
Moderate 3/5	120 to 220 kWh/m2	Some winter sun.
Good 4/5	220 to 320 kWh/m2	Good winter sun.
High 5/5	320 or more kWh/m2	A lot of free winter heat.

This house: North wall 356 High 5/5 | East wall 204 Moderate 3/5 | South wall 86 Low 2/5 | West wall 191 Moderate 3/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3).

Wall heat transmittance U (NBR 15575-4) Brazilian norm

How fast heat flows through the wall (lower = better insulated). Limit in zones 3-8: 3.7 if the outside paint is light (solar absorptance up to 0.6), 2.5 if darker.

Meets both limits 5/5	below 2.5 W/m2K	Meets the limit even with dark paint.
Light paint only 4/5	2.5 to 3.7 W/m2K	Meets the limit only with light-coloured paint (absorptance 0.6 or less).
Fails 1/5	3.7 or more W/m2K	Above the limit: the simplified method is not met.

This house: External wall heat transmittance U 2.52 (limit <= 3.7) Light paint only 4/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Wall heat capacity (NBR 15575-4) Brazilian norm

How much heat the wall stores (higher = calmer indoor temperature). Limit: at least 130 in zones 1-7.

Fails 1/5	below 130 kJ/m2K	Less heat storage than the minimum.
Meets 5/5	130 or more kJ/m2K	Meets the minimum of 130.

This house: External wall heat capacity 134 (limit >= 130) Meets 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Roof heat transmittance U (NBR 15575-5) Brazilian norm

Heat flow through roof + ceiling. Limit in zones 3-6: 2.3 if the roof is light-coloured (absorptance up to 0.6), 1.5 if darker (e.g. red clay tile).

Meets both limits 5/5	below 1.5 W/m2K	Meets the limit even for a dark roof.
Light roof only 4/5	1.5 to 2.3 W/m2K	Meets the limit only if the roof surface is light.
Fails 1/5	2.3 or more W/m2K	Above the limit.

This house: Roof heat transmittance U (indicative) 1.78 (limit <= 1.5) Light roof only 4/5

Source: ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs. Our roof U is indicative: a ventilated attic is not modelled by the standard calculation.

6.4 NBR 15575-1:2021 thermal performance by simulation (PHFT)

Calculated with: **EnergyPlus + AirflowNetwork** + **OpenFOAM (CFD)** · EnergyPlus solves the heat and air flow, OpenFOAM gives the wind pressure on the openings · details and reliability in section 9

The 2021 edition of NBR 15575-1 judges the thermal performance of a dwelling by computer simulation: the house as designed (**real model**) is compared with the same house built with reference materials (**reference model**: 100 mm walls and floors, tile + air chamber + 100 mm slab roof, windows of 17% of the floor area, no fixed shading). The main indicator is the **PHFT**, the percentage of occupied hours with an operative temperature between 18 and 26 °C (the range for climates with an annual mean below 25 °C: this weather file has 23.0 °C), over the bedrooms and the living room, with windows opened only while the room is occupied and the indoor air is at least 19 °C and warmer than outdoors. The standard also limits the hottest and coldest operative temperature and, for the intermediate and superior levels, asks for a reduction of the annual heating and cooling load (ideal air conditioner at 23 °C / 21 °C while occupied).

Room	PHFT real	PHFT reference	Hot hours %	Cold hours %	T_{max} / T_{min} real
Living + dining	42.4%	34.9%	56.3	1.3	35.1 / 15.9 °C
Bedroom 1	64.7%	66.5%	32.7	2.5	32.7 / 15.2 °C
Bedroom 2	59.9%	63.1%	38.1	2.0	33.2 / 16.0 °C
Bedroom 3	56.5%	60.1%	41.2	2.4	33.2 / 15.7 °C

Result for the house as it is: PHFT of the dwelling = 55.9% against 56.1% of the reference (minimum level needs more than 90% of the reference: **yes**). Hottest operative temperature 35.1 °C against 36.7 °C allowed up to +2 °C: **yes**. Coldest 15.2 °C against 13.6 °C allowed down to −1 °C: **yes**. Total load 6,984 kWh/yr against 8,641 kWh/yr of the reference (19.2% lower). Level reached: **Minimum**.

Package	PHFT real	ΔPHFT (needed 12.4)	T_{max}	T_{min}	Load reduction	Level
Base	55.9%	-0.3	35.1	15.2	19.2%	**Minimum**
Rock wool 50 mm	67.1%	11.0	31.7	16.6	52.0%	**Minimum**
Glass wool 50 mm	67.6%	11.4	31.6	16.6	53.0%	**Minimum**
PET wool 50 mm	67.6%	11.4	31.6	16.6	53.0%	**Minimum**
Sheep wool 50 mm	67.6%	11.4	31.6	16.6	53.0%	**Minimum**
Rock wool 100 mm	69.7%	13.5	31.1	16.8	57.6%	**Superior**
Draught sealing	55.9%	-0.3	35.1	15.2	19.2%	**Minimum**
Double glazing	55.9%	-0.3	35.0	15.4	19.8%	**Minimum**
Low-E double glazing	56.9%	0.8	34.8	15.3	23.5%	**Minimum**
Fixed brise 0.6 m	56.7%	0.6	35.0	15.2	21.4%	**Minimum**
Movable blinds	56.9%	0.8	35.0	15.1	22.3%	**Minimum**
All together	71.0%	14.8	31.0	16.8	60.2%	**Superior**

The intermediate level needs ΔPHFT of at least 45 − 0.58 × PHFT of the reference (single-family house) and the load reduction of table 20; the superior level needs the same ΔPHFT and a reduction of 35% (reference load below 100 kWh/m²·yr) or 55% (above), or a PHFT of 95% or more. Packages with draught sealing, glazing and shading barely move the indicators; roof insulation is what moves the house towards the higher levels (needed ΔPHFT 12.4; 50 mm wools reach 11.4). Levels reached: Rock wool 100 mm (Superior), All together (Superior). A package that lands exactly on the needed value is borderline: the result moves with small changes of the model.

The levels of the standard and what they mean for living in the house

Level	What the standard asks	What living in the house is like	In this house
Not met	The PHFT is 90% of the reference house or less, or the hottest / coldest operative temperature is more than 2 °C / 1 °C worse than the reference.	The house is clearly worse than the standard reference: more hot afternoons and cold nights indoors, and the air conditioner has to carry the comfort most of the year.	No package reaches it.
Minimum	PHFT above 90% of the reference house; hottest operative temperature at most 2 °C above the reference and coldest at most 1 °C below it.	About as comfortable as a house built with the reference materials (100 mm walls, tile and slab roof). On hot afternoons and cold dawns the rooms leave the 18-26 °C band; expect to use the fan or the air conditioner on many days and to wear a jacket on cold mornings.	**Current house**: PHFT 55.9% against 56.1% of the reference (ΔPHFT -0.3), load reduction 19.2%.
Intermediate	Everything of the minimum level, plus a PHFT gain over the reference house (ΔPHFT) of at least 45 − 0.58 × PHFT of the reference (single-family house) and a reduction of the annual heating and cooling load (table 20).	Noticeably more hours inside the comfortable band without any equipment, rooms that stay cooler in the afternoon and warmer at dawn, and a smaller air-conditioning bill. The air conditioner becomes a help for the hottest days, not a daily need.	No package reaches it.
Superior	Everything of the intermediate level, with a load reduction of 35% (reference load below 100 kWh/m²·year) or 55% (above), or a PHFT of 95% or more.	The house holds a comfortable temperature almost by itself: few hot or cold hours, the air conditioner is rarely needed and, when it is, a small unit is enough. This is the level that pays back in the bill.	Reached by **All together**: PHFT 71.0%, ΔPHFT 14.8, load reduction 60.2%.

The 'In this house' column uses the results of this simulation; the description of life in the house is our reading of the numbers, not text of the standard.

Cross ventilation (wind-driven)

Natural ventilation is solved by the EnergyPlus AirflowNetwork: every window, open door and crack is a flow path whose pressure comes from the wind (pressure coefficients of every opening, calculated by the OpenFOAM CFD for 24 wind directions), from the temperature difference and from the height of the opening. Air changes per hour through the exterior windows and doors, base house:

Calculated with: **EnergyPlus** + **AirflowNetwork** + **OpenFOAM (CFD)** · air flow by the AirflowNetwork with CFD pressure coefficients · details and reliability in section 9

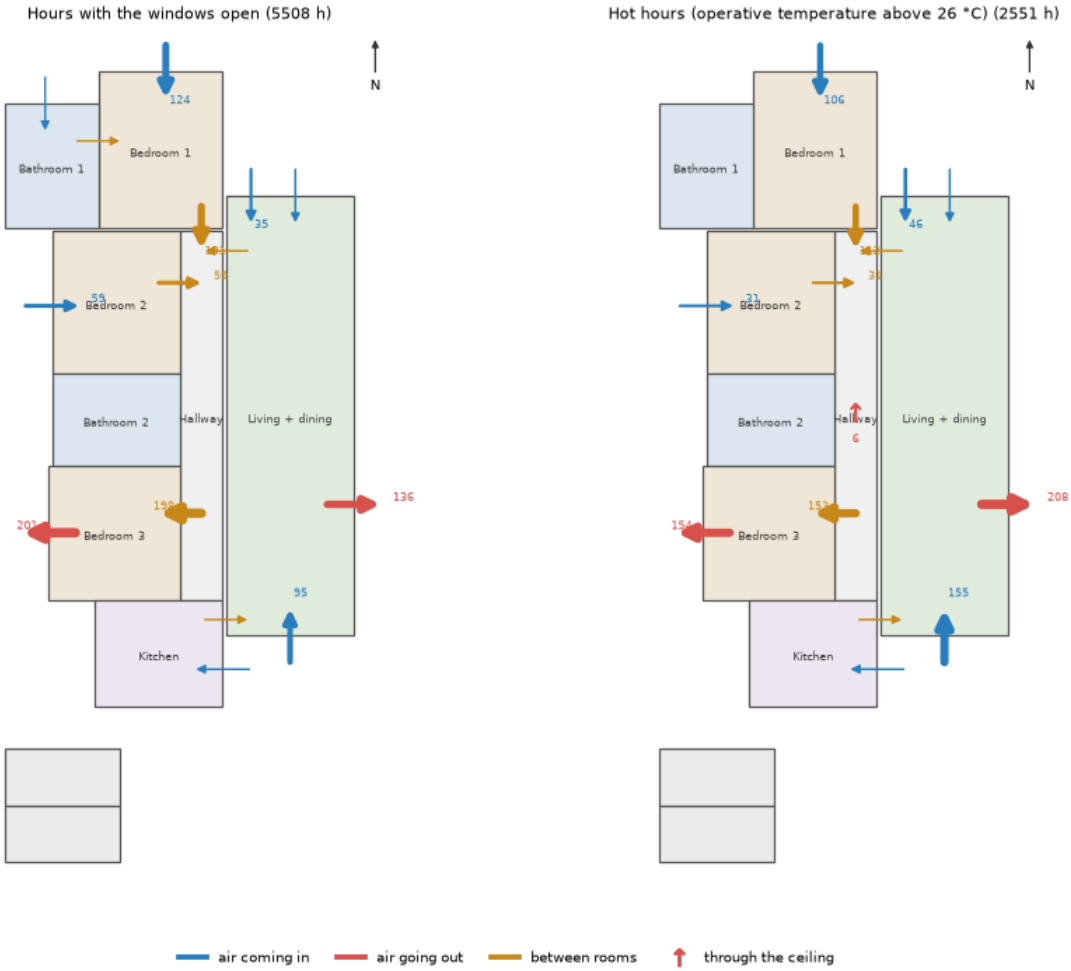
Room	Mean ACH	90th percentile	Hours above 5 ACH
Living + dining	2.2	8.8	1,742
Bedroom 1	2.5	6.0	2,249
Bathroom 1	1.2	1.7	0
Bedroom 2	2.4	6.5	2,705
Bathroom 2	1.3	1.8	0
Bedroom 3	2.9	8.0	3,077
Kitchen	0.3	0.6	0

How the air moves through the house

Calculated with: **EnergyPlus** + **AirflowNetwork** + **OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 9

The airflow network gives, for every hour, the volume of air that crosses each window, door and opening. The figure shows the average of those flows on the plan, for all the hours with the windows open and for the hot hours (operative temperature above 26 °C); arrows are blue where fresh air comes in, red where air leaves the house, orange for air moving between rooms, and the thickness follows the flow in m³/h. This is the flow between rooms (one air node per room), not a velocity field inside each room: it shows which openings feed the house and where the air leaves, not the draught at a given spot.

How the air moves through the house (net flow of every opening) — m³/h



Opening	Type	Air goes from	to	m³/h
window 2 · Living + dining	window	Living + dining	outdoors	208
window 3 · Living + dining	window	outdoors	Living + dining	155
window 1 · Bedroom 3	window	Bedroom 3	outdoors	154
door 5 · Hallway	door	Hallway	Bedroom 3	153
door 2 · Bedroom 1	door	Bedroom 1	Hallway	112
window 1 · Bedroom 1	window	outdoors	Bedroom 1	106
window 1 · Living + dining	window	outdoors	Living + dining	46
window 1 · Bedroom 2	window	outdoors	Bedroom 2	31
door 4 · Hallway	door	Bedroom 2	Hallway	31

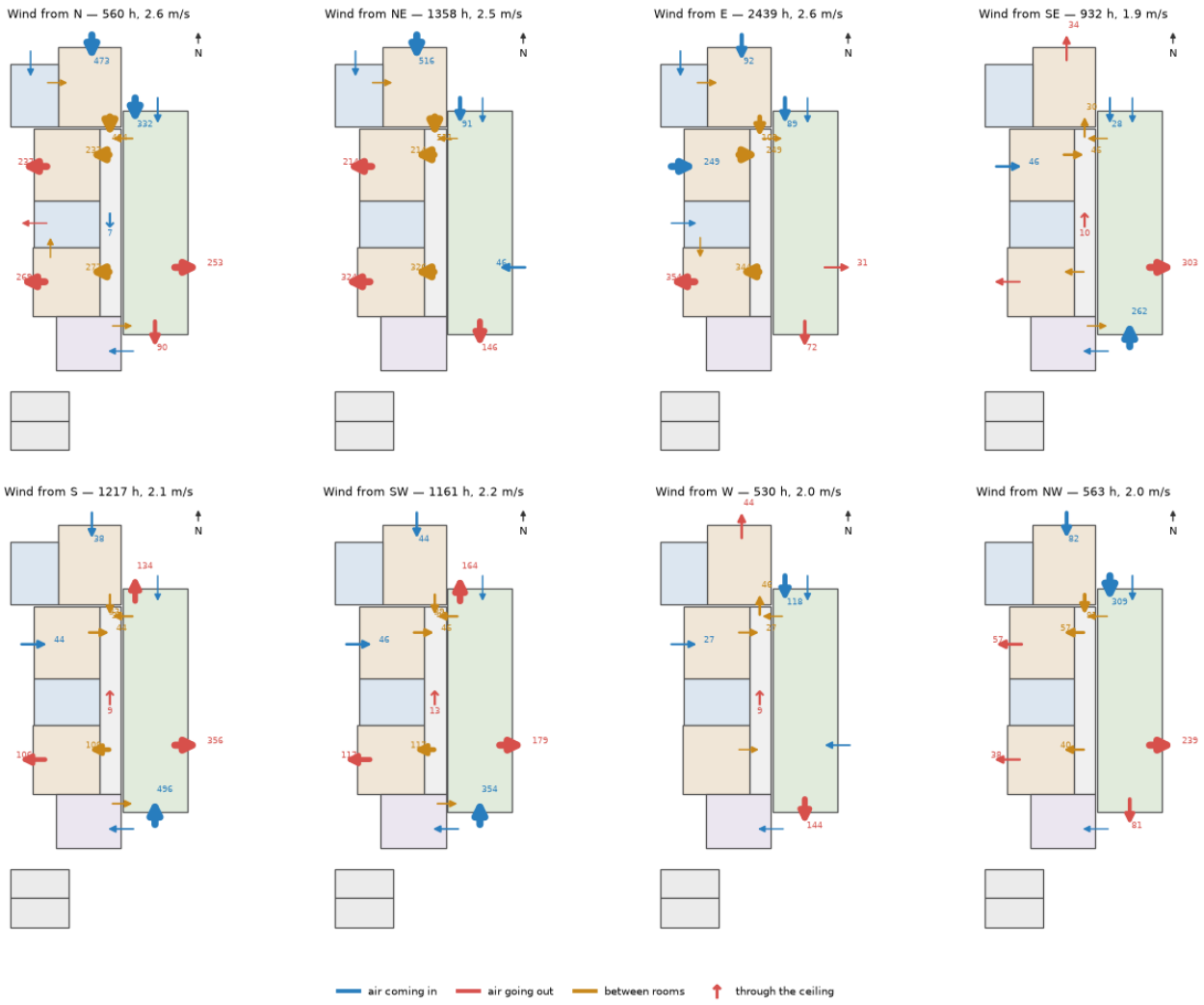
Main air paths of the house as it is in the hot hours (flows below 20 m³/h omitted).

Circulation for each wind direction

> Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 9

Same flows, separated by the direction the wind comes from (eight sectors of 45°, wind of the weather file). The pressure on each opening comes from the 24 CFD cases, so the pattern changes with the wind: openings on the windward side feed the house and those on the leeward side or under suction let the air out.

Air flow for each wind direction (hours with the windows open) — m³/h

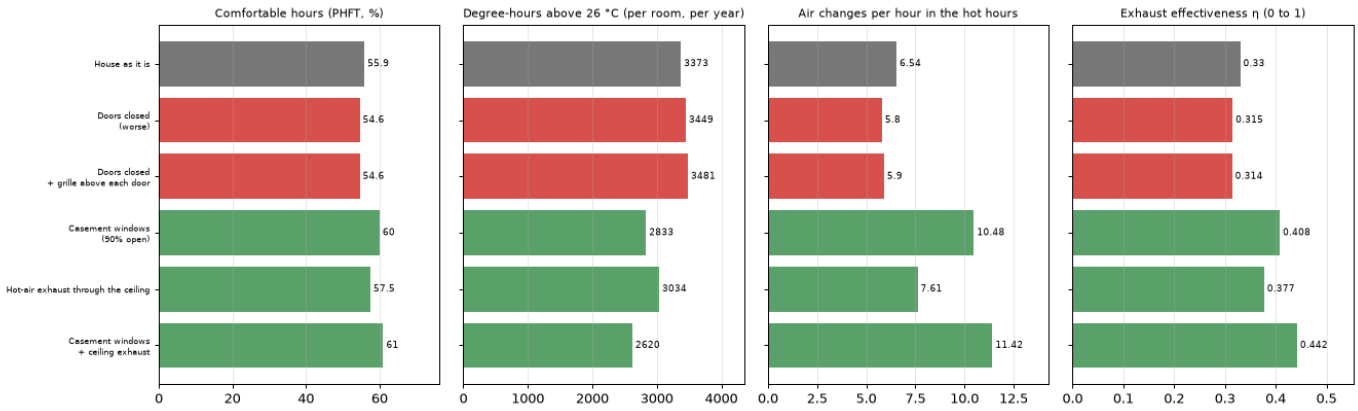


Plan changes that improve or worsen ventilation

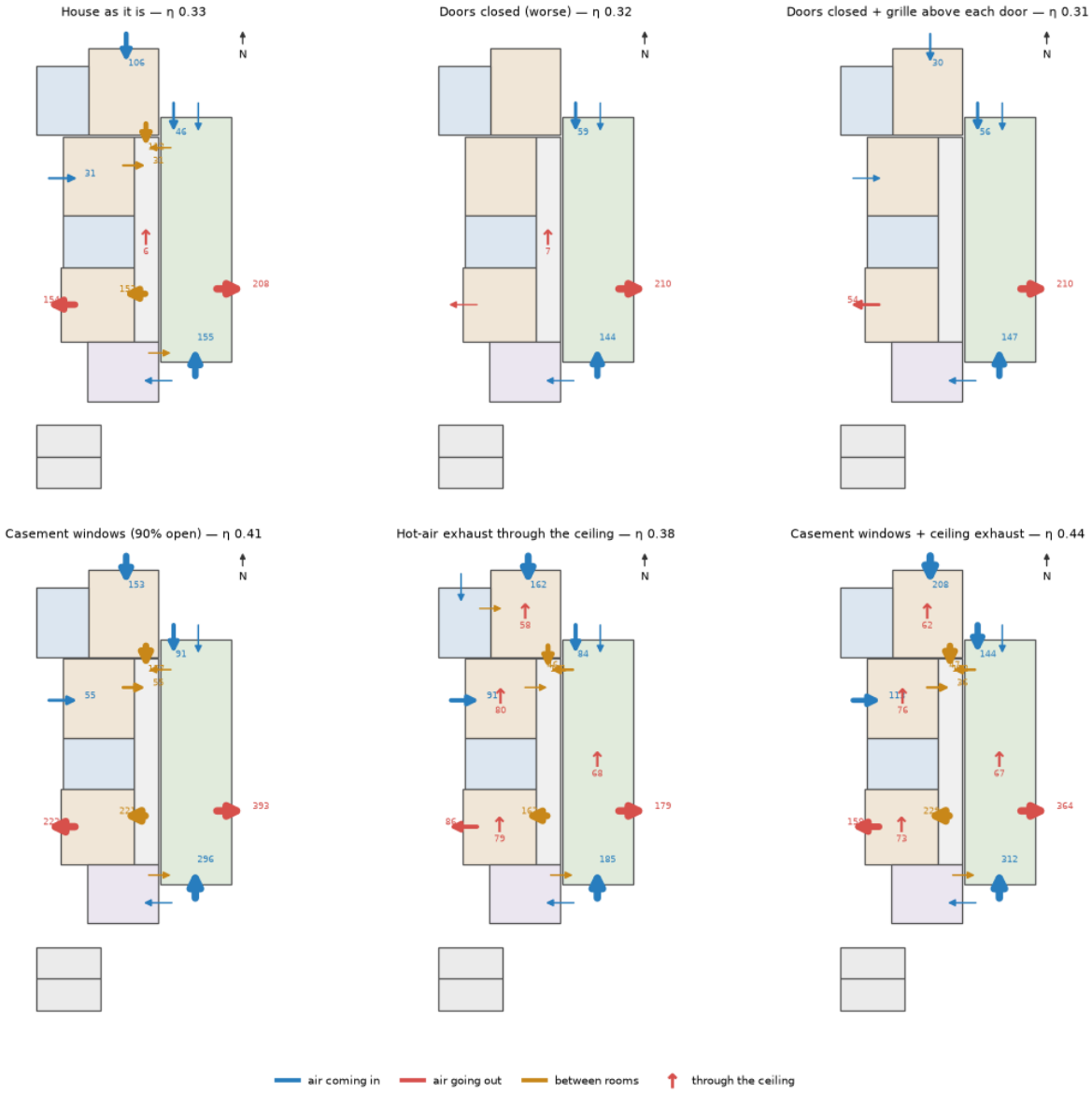
Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 9

Each change is a new full-year run of the same model, with the same weather, occupancy and CFD pressures; only the opening in question changes. The table compares comfort and the exhaust of hot air, using the occupied hours of the bedrooms and the living room.

Change	PHFT	Hottest °C	Degree-hours above 26 °C	Air changes per hour in hot hours	Exhaust effectiveness η	Heat carried out (kWh/year)
House as it is Sliding windows (50% open), doors of the bedrooms and the living room open while the room is occupied.	55.9%	35.1	3,373	6.5	0.33	7,541
Doors closed (worse) The doors between the bedrooms / living room and the rest of the house stay closed: air has no way through.	54.6% (-1.3)	35.2	3,449 (+76)	5.8 (-0.7)	0.32 (-0.02)	6,618 (-923)
Doors closed + grille above each door Same closed doors, with a permanent 0.5 x 0.4 m grille above each one (the usual Brazilian transom grille).	54.6% (-1.3)	35.1	3,481 (+108)	5.9 (-0.6)	0.31 (-0.02)	6,854 (-687)
Casement windows (90% open) Windows that open 90% of the glazed area (casement or tilt) instead of the 50% of sliding windows; same size and frames.	60.0% (+4.1)	34.9	2,833 (-540)	10.5 (+3.9)	0.41 (+0.08)	9,005 (+1464)
Hot-air exhaust through the ceiling A 0.2 m2 grille in the ceiling of each bedroom and of the living room, and eave openings of the attic 5 times larger (2% of the floor area instead of 0.4%).	57.5% (+1.6)	35.0	3,034 (-339)	7.6 (+1.1)	0.38 (+0.05)	11,180 (+3639)
Casement windows + ceiling exhaust Casement windows (90% open) together with the ceiling grilles and the larger eave openings.	61.0% (+5.1)	35.0	2,620 (-753)	11.4 (+4.9)	0.44 (+0.11)	12,533 (+4992)



Air flow in the hot hours for each floor-plan change — m³/h



Reading: the best change for comfort is **Casement windows + ceiling exhaust** (PHFT 61.0% against 55.9%, exhaust effectiveness 0.44 against 0.33); the worst is **Doors closed (worse)** (PHFT 54.6%). The reference house has PHFT 56.1%.

Hot-air exhaust effectiveness

> Calculated with: **EnergyPlus + AirflowNetwork** · temperatures and flows of the airflow network · details and reliability in section 9

There is no single standard coefficient for this, so the report uses three measurable indicators: (1) the **air changes per hour in the hot hours**; (2) the **exhaust effectiveness η** = (T closed − T ventilated) ÷ (T closed − T outdoor), computed with the operative temperatures of each room in the occupied hours when the house with the windows closed is above 26 °C and more than 1 °C above the outdoor air (0 = the ventilation does not cool the room, 1 = the room follows the outdoor air); (3) the **heat carried out** by the air that leaves the house, ρ·cp·V·(T zone − T outdoor), in kWh per year. Age of air and the removal of a plume of hot air at ceiling level need an interior CFD, which this study does not do.

The wind pressure coefficients (Cp) of every window, door and eave opening come from a **CFD simulation with OpenFOAM** (steady RANS, k-epsilon, atmospheric boundary layer over suburban roughness, 24 wind directions (every 15°), 1.2 to 2.0 million cells each; pressure read 0.3 m in front of each opening and expressed as a coefficient of the dynamic pressure of the EnergyPlus wind profile at the height of the opening). Against the generic Swami & Chandra coefficients the CFD gives stronger suction and overpressure at the openings (largest windward Cp 0.37 against 0.60; mean difference per opening 0.42). Effect on the result: PHFT of the dwelling 55.9% with CFD against 57.5% with the generic coefficients, reference 56.1% against 57.0% (ΔPHFT -0.3 against 0.5): the verdict does not depend on the choice.

The house in the CFD follows the real geometry of the IFC model: the two roofs with their valley and ridge (up to 4.1 m), the eaves and the carport and porch roofs as thin plates, with windows and doors closed. A first version used rooms extruded to a flat 3.5 m top; the real roof shape changes the coefficients by 0.63 on average per opening (root mean square over the 12 wind directions both versions have), mostly at the attic openings and the windows under the eaves.

How far the CFD can be trusted (numerical verification)

- **Box check:** a plain 22 × 11 × 3 m box gives a mean Cp of 0.69 on the windward wall and -0.35 on the leeward wall; the Eurocode (EN 1991-1-4, table 7.1, zones D and E, h/d = 0.27) gives 0.70 and -0.31.
- **Mesh:** three meshes (cell 2.0 / 1.2 / 0.8 m, up to 1.1 million cells) differ by an rms of 0.06 (coarse vs medium) and 0.07 (medium vs fine) in Cp over 64 opening-direction points: the differences do not shrink with refinement, so the mesh error is about ±0.1 and no convergence order can be claimed.
- **Terrain roughness** (0.1 m and 1.0 m instead of 0.3 m): rms 0.11 and 0.14. **Domain** 1.5 times larger: rms 0.04.
- **Combined uncertainty of each opening's Cp:** median ±0.21, 95% of the openings within ±0.43. Moving every coefficient by this amount at once changes the PHFT of the dwelling by 0.1 point(s) and of the reference by 0.1 point(s): the NBR 15575 result does not depend on this uncertainty.
- **Turbulence model:** the standard k-epsilon model was kept: on the plain box its mean Cp is closest to the Eurocode (windward 0.69, error 0.05) against realizable k-epsilon (0.65, 0.08) and k-omega SST (0.57, 0.14); on the house the models differ by an rms of 0.02 (realizable) and 0.08 (SST), which is part of the uncertainty above.
- **Wind profile matching:** EnergyPlus applies Cp to the dynamic pressure of its own wind profile (power law) at the opening height, not to the CFD's. The CFD pressures were converted accordingly (factor 0.4 to 0.8 depending on height); without it the wind pressure would be overstated by up to 2.5 times at the windows.
- Wall-function values y+ are large (1,000 to 4,000), as usual for full-scale buildings; the mean pressures are not very sensitive to this, but the wall shear is not reliable.

CFD limits: the lot walls and the neighbouring buildings are included but their heights are estimates from photographs, and there are no trees; windows and doors are closed in the CFD (only the outside is solved); a steady RANS model gives mean pressures only (no gusts, no unsteady wake). Use it as a better-than-generic estimate, not a wind-tunnel result.

Source of the procedure: section 11.4 of ABNT NBR 15575-1:2021 (full text, copy kept in norms/source_docs) and the public amendment draft of March 2021 (LabEEE); tables 3, 4, 20 and 21 agree in both, the equation of figure 4 (ΔPHFTmín = 45 − 0.58 × PHFT of the reference, single-family) was read from the figure of the draft. Bioclimatic zone: a non-official article (UGREEN, October 2026, kept in norms/source_docs) says Maringá moved from zone 1 to zone 4B in NBR 15220-3:2024, which is among the zones where the minimum-temperature criterion applies; the 2024 standard itself was not checked. Assumptions of ours: the kitchen and the hallway are transient rooms (not counted in the PHFT); the glass sliding door is a window; interior doors between a bedroom or the living room and a transient room stay open; the 18-26 °C band and the 23/21 °C set points apply because the annual mean temperature is below 25 °C;the wind pressure coefficients come from the CFD (the generic Swami & Chandra formula is kept only as a comparison); the lot walls and neighbouring buildings are in the CFD with estimated heights but not in the shadow and daylight models; ground temperature is the annual mean of the weather file.

6.5 Wind CFD: how it was calculated and how far it can be trusted

Calculated with: **OpenFOAM (CFD)** · details and reliability in section 9

This section details the wind simulation behind the cross-ventilation results above. In short: the house is placed in a virtual wind tunnel, the air flow around it is solved once for each wind direction, and the pressure the wind exerts in front of every window, door and eave opening is turned into a coefficient (Cp) that the EnergyPlus airflow network multiplies by the real wind speed of each hour.

Calculation setup

Wind directions	24 (every 15°)
House geometry	closed envelope rebuilt from the IFC model: the two roofs with valley and ridge (up to 4.1 m), the eaves, the carport and porch roofs as thin plates; windows and doors closed (only the outside is solved)
Computational domain	6H upstream, 15H downstream, 5H to each side and 6H above, with H the building height (4.1 m): the usual rules of computational wind engineering
Mesh	snappyHexMesh (castellated + snap); background cells of 1.2 m refined to about 0.15 m on the house; 1,154 to 1,991 thousand cells per direction
Approach wind	atmospheric boundary layer with a logarithmic profile, 10 m/s at 10 m over suburban terrain (roughness length 0.3 m)
Equations and model	steady Reynolds-averaged Navier-Stokes with the kEpsilon turbulence model, solved by simpleFoam (OpenFOAM v2312, Docker); wall functions on the house and on the rough ground
Convergence	residuals below p 1e-5, U 1e-6, k/epsilon 1e-5; median of 344 iterations (maximum 885)
Surroundings	lot walls (3.2 m on the east side up to the carport and 2.6 m after it, 1.8 m on the west and at the back, 3.2 m where the back wall meets the neighbour's edicula) and the neighbouring houses, the houses across the street and the neighbour's edicula as closed boxes. All heights are estimates from photographs and the owner's description (single-storey neighbours with a 4.4 m ridge, 1.5 m from the wall); they are meshed more coarsely (0.6 m) than the house
Where pressure is read	34 points, each 0.3 m in front of an exterior window, door, wall crack or attic vent; mean of the last 20 iterations

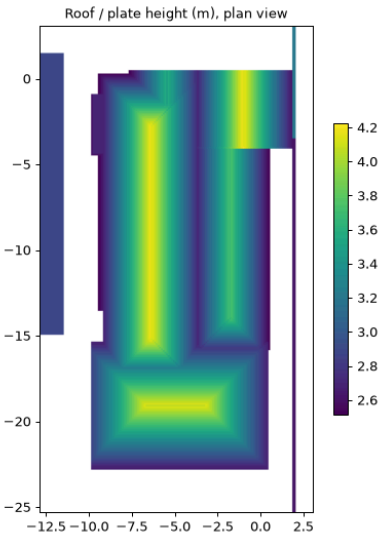
From pressure to the coefficient used by EnergyPlus

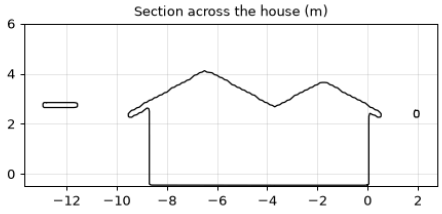
EnergyPlus computes the wind pressure on an opening as $Cp \times \frac{1}{2}\rho U(z)^2$, with U(z) the wind speed at the opening height taken from its own profile (power law with exponent 0.22 for suburbs). The CFD gives the real pressure p, so the coefficient is $Cp = p / (\frac{1}{2}\, U_EP(z)^2)$. Because the CFD inlet is logarithmic, this is the same as multiplying the CFD coefficient by $(U_CFD(z) / U_EP(z))^2$: 0.39 / 0.49 / 0.61 / 0.79 at 0.75 / 1.05 / 1.6 / 3.05 m. Without this step the wind pressure on the windows would be overstated by up to 2.5 times.

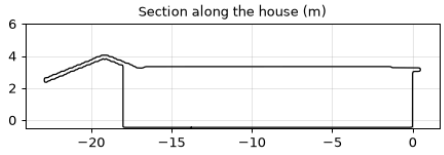
Results of the numerical verification

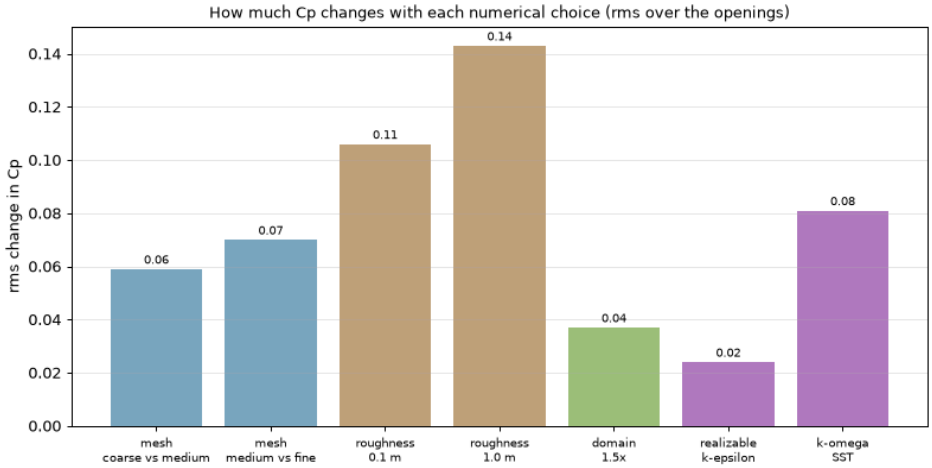
Check	How	Result (rms change in Cp)
Mesh	cells of 2.0 / 1.2 / 0.8 m (94 thousand to 1.1 million cells), two wind directions	0.06 / 0.07; does not shrink with refinement
Terrain roughness	0.1 m and 1.0 m instead of 0.3 m	0.11 / 0.14
Domain size	1.5 times larger in every direction	0.04
Turbulence model	realizable k-epsilon and k-omega SST instead of standard k-epsilon	0.02 / 0.08
Reference case	plain 22 × 11 × 3 m box against the Eurocode (EN 1991-1-4, table 7.1, zones D and E)	windward 0.69 vs 0.70; leeward -0.35 vs -0.31
Generic coefficients	Swami & Chandra, used when no CFD exists	0.42
First version of the CFD	rooms extruded to a flat 3.5 m top	0.63

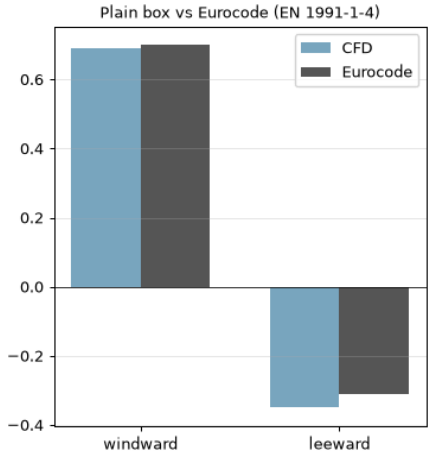
Combining the mesh, roughness, domain and turbulence-model differences gives an uncertainty of ±0.21 (median) and ±0.43 (95% of the openings) in each Cp. Moving every coefficient up and down by that amount at once changes the PHFT of the dwelling between 55.4% and 55.3% and that of the reference between 53.8% and 53.7%: the NBR 15575 level does not depend on it.

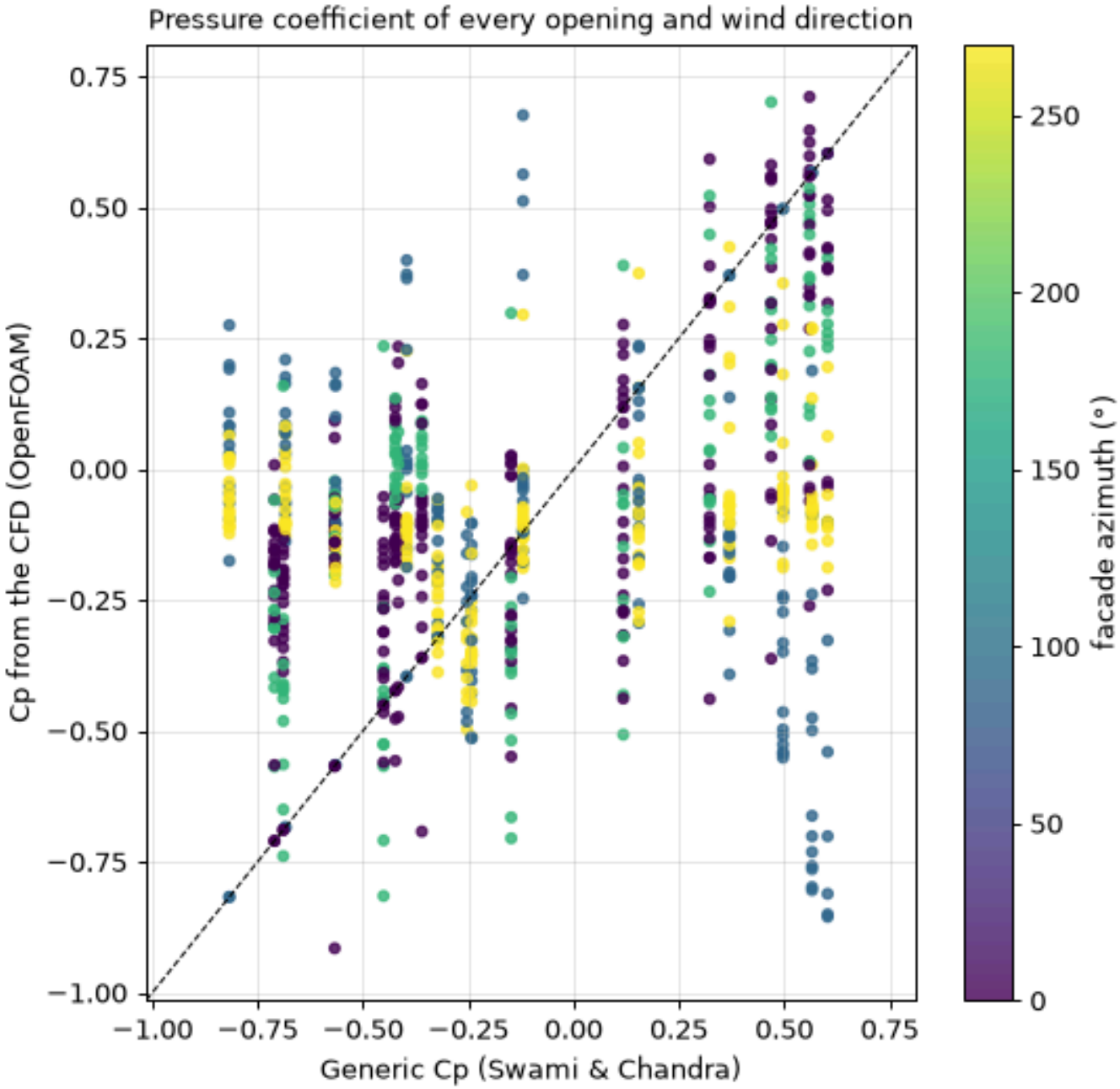

Roof / plate height (m), plan view
0
−5
−10
−15
−20
−25
−12.5 −10.0 −7.5 −5.0 −2.5 0.0 2.5
4.2
4.0
3.8
3.6
3.4
3.2
3.0
2.8
2.6


Section across the house (m)
6
4
2
0
−12 −10 −8 −6 −4 −2 0 2
0


Section along the house (m)
6
4
2
0
−20 −15 −10 −5 0


How much Cp changes with each numerical choice (rms over the openings)
rms change in Cp
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.06
0.07
0.11
0.14
0.04
0.02
0.08
mesh
coarse vs medium
mesh
medium vs fine
roughness
0.1 m
roughness
1.0 m
domain
1.5x
realizable
k-epsilon
k-omega
SST


Plain box vs Eurocode (EN 1991-1-4)
CFD
Eurocode
0.6
0.4
0.2
0.0
−0.2
−0.4
windward
leeward



Figures: the house as the CFD sees it (plan of the roof heights and two sections); how much Cp changes with each numerical choice and the box check; and the CFD coefficient of every opening and wind direction against the generic ones (points on the dashed line agree).

7. Upgrades: insulation, sealing, glazing and shading

Each upgrade below is simulated on the whole year and compared with the house as it is today. **Roof wool** lies on top of the ceiling (attic side); **sealing** means rubber gaskets and brush seals on the frames and doors; **double glazing** replaces the single glass; a **fixed brise** is a horizontal projection above each window; **movable blinds** close by themselves in the model when the sun on the window is strong. The last row combines the main measures.

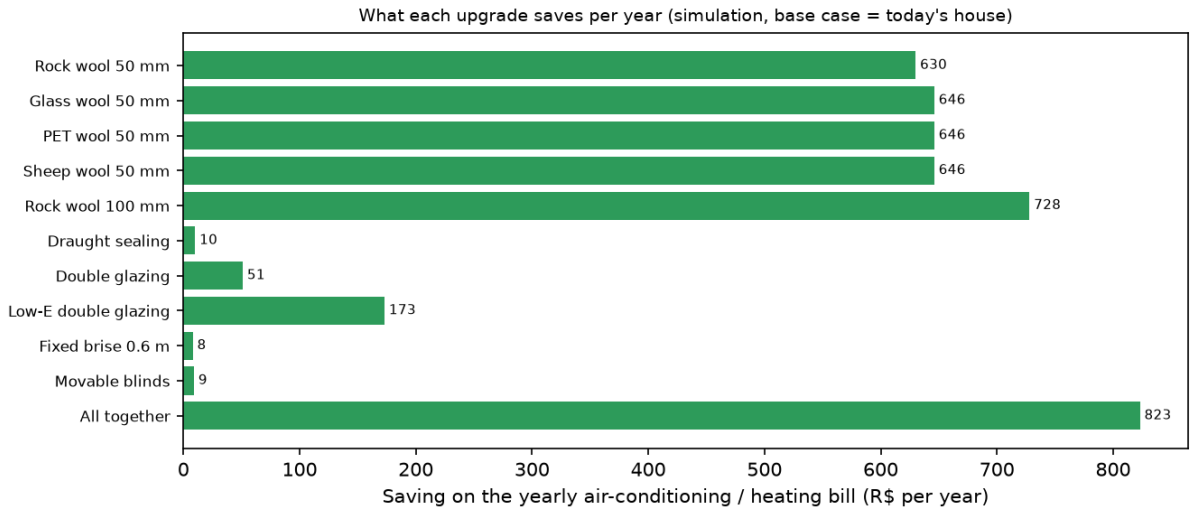
7.1 Results by upgrade

Upgrade	What it is	Comfortable hours	Hours too hot	Hottest hour	Cooling R$/yr	Change in bill	Daylight
Base	Existing house	83.9% Good 4/5	554 h	32.3 °C	R$ 1,582	R$ 0/yr	100%
Rock wool 50 mm	Rock wool 50 mm	88.9% Good 4/5	311 h	30.6 °C	R$ 952	R$ -630/yr	100%
Glass wool 50 mm	Glass wool 50 mm	89.0% Good 4/5	306 h	30.6 °C	R$ 936	R$ -646/yr	100%
PET wool 50 mm	PET wool (recycled bottles) 50 mm	89.0% Good 4/5	307 h	30.6 °C	R$ 936	R$ -646/yr	100%
Sheep wool 50 mm	Sheep wool 50 mm	89.0% Good 4/5	307 h	30.6 °C	R$ 936	R$ -646/yr	100%
Rock wool 100 mm	Rock wool 100 mm	89.4% Good 4/5	286 h	30.3 °C	R$ 854	R$ -728/yr	100%
Draught sealing	Rubber gaskets + brush seals on frames and doors	84.0% Good 4/5	551 h	32.3 °C	R$ 1,572	R$ -10/yr	100%
Double glazing	Double glazing, clear	84.2% Good 4/5	548 h	32.3 °C	R$ 1,531	R$ -51/yr	89%
Low-E double glazing	Double glazing, low-E	84.5% Good 4/5	513 h	32.2 °C	R$ 1,409	R$ -173/yr	80%
Fixed brise 0.6 m	Fixed horizontal brise / overhang 0.6 m	83.9% Good 4/5	548 h	32.3 °C	R$ 1,574	R$ -8/yr	100%
Movable blinds	Movable blinds / shutters (closed when the sun is strong)	83.9% Good 4/5	547 h	32.3 °C	R$ 1,573	R$ -9/yr	100%
All together	Rock wool 50 mm; Rubber gaskets + brush seals on frames and doors; Double glazing, low-E; Fixed horizontal brise / overhang 0.6 m; Movable blinds / shutters (closed when the sun is strong)	89.5% Good 4/5	284 h	30.4 °C	R$ 759	R$ -823/yr	80%

Calculated with: **Fast engine (Python, this project)** · EnergyPlus cross-check in 7.4 · details and reliability in section 9

Where the numbers of the table above come from:

Column	How it is calculated	Engine	How far to trust it
Comfortable hours	Share of the 8,760 hours of the year in which the operative temperature of the room (mean of air and wall-surface temperature) is inside the 80% acceptability band of the ASHRAE 55 adaptive model; average of the rooms. The air conditioner runs on its schedule in this calculation, so the percentage is higher than the PHFT of section 6.4 (house without air conditioning).	Fast engine. Cross-checked with EnergyPlus in 7.4.	Medium: good for ranking the upgrades, the absolute percentage is approximate.
Hours too hot	Hours of the year with the operative temperature above the upper limit of the same band; average per room.	Fast engine.	Medium, same as above.
Hottest hour	Highest operative temperature of the year in each room, averaged over the rooms. It comes from one hour of the typical weather year, so real heat waves will be hotter.	Fast engine.	Medium-low: a single extreme hour is sensitive to the weather file.
Cooling R$/yr	Heat removed by an air conditioner that holds 26 °C in the hours each room kind is used, divided by the efficiency (COP 3.0) and multiplied by the tariff (R$ 1.00/kWh, an assumption set in the project file).	Fast engine (energy); tariff and COP are assumptions.	Low: use it as an order of magnitude and to compare upgrades. EnergyPlus gives roughly half of this energy (7.4).
Change in bill	Cooling (and heating, when enabled) cost of the upgrade minus the cost of the house as it is. Negative = saving.	Fast engine.	Low-medium: the difference between upgrades is more reliable than the absolute value.
Daylight	Visible transmittance of the new glass divided by that of the existing glass. Only the glass is counted: blinds and shading are not applied here, and no light engine is run for this column.	Catalogue value (no engine); the daylight of the house itself is in the daylight sections (Radiance).	Medium: a catalogue value, not a calculation.



7.2 Which wool and how thick? (change in yearly bill and average comfortable hours)

Wool	25 mm	50 mm	75 mm	100 mm	150 mm
Glass wool λ 0.040 W/m·K	R$ -516 88.1%	R$ -646 89.0%	R$ -705 89.3%	R$ -738 89.5%	R$ -774 89.7%
Rock wool λ 0.044 W/m·K	R$ -496 88.0%	R$ -630 88.9%	R$ -693 89.2%	R$ -728 89.4%	R$ -766 89.6%
PET wool (recycled bottles) λ 0.040 W/m·K	R$ -516 88.1%	R$ -646 89.0%	R$ -705 89.3%	R$ -738 89.5%	R$ -774 89.6%
Sheep wool λ 0.040 W/m·K	R$ -517 88.1%	R$ -646 89.0%	R$ -705 89.3%	R$ -738 89.5%	R$ -773 89.6%

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 9

Negative = saving per year. More wool always helps, with diminishing returns: the first 50 mm give most of the benefit. Wools differ little in conductivity; the choice is usually decided by price, fire behaviour, handling and humidity.

7.3 Values used and how reliable they are

Measure	Value used	Reliability of the value	Source
Glass wool	λ 0.040 W/m·K, ρ 15 kg/m³	typical	range 0.030-0.046 W/mK (Saint-Gobain, wool insulation comparison); mid-range value
Rock wool	λ 0.044 W/m·K, ρ 40 kg/m³	verified	Rockwool datasheet (panel, 0.044 W/mK at 50 C, EN 12667)
PET wool (recycled bottles)	λ 0.040 W/m·K, ρ 30 kg/m³	assumption	typical manufacturer range 0.037-0.041 W/mK; check the product you buy
Sheep wool	λ 0.040 W/m·K, ρ 25 kg/m³	assumption	typical range 0.038-0.042 W/mK; check the product you buy
Single glass (existing)	U 5.8 W/m²K, g 0.85, VT 0.88	verified	5.8 W/m2K is the single-glass limit of EN 673 / ISO 10077
Double glazing, clear	U 2.8 W/m²K, g 0.76, VT 0.78	typical	typical catalogue values for 4-12-4 clear double glazing (EN 673 / NFRC ranges)
Double glazing, low-E	U 1.8 W/m²K, g 0.40, VT 0.70	typical	typical catalogue values for low-E (solar control) double glazing
Rubber gaskets + brush seals on frames and doors	closed-house air changes −25%	assumption	AIVC: weatherstripping alone cuts house air change 10-14.7%, a full retrofit up to 27.5%; 25% used for frames and doors
Fixed horizontal brise / overhang	depth set per package	assumption	analytic profile-angle shading of the window (sun-path geometry, pvlib); depth in metres from project.json
Movable blinds / shutters (closed when the sun is strong)	closed gain x0.3	assumption	external blinds let through about 10-30% of solar gain when closed; closed when direct sun on the window exceeds 250 W/m2 (08-18 h)

Calculated with: **3D model (IFC) and tables** · catalogue values, not calculated · details and reliability in section 9

The upgrade prices are not included: with a quotation you can divide it by the yearly saving to get the payback time. Daylight column: share of the daylight that still enters (glass transmittance only; blinds and brise are not applied to daylight).

7.4 Cross-check with a reference engine (EnergyPlus)

Calculated with: **EnergyPlus + AirflowNetwork** · details and reliability in section 9

Every package above was recalculated with **EnergyPlus** (US Department of Energy), the open-source engine validated by ASHRAE Standard 140 and used inside DesignBuilder and OpenStudio. It solves the heat balance of every surface with conduction transfer functions, shares walls between rooms, models the attic above each room, shades windows with the real eaves and uses wind- and buoyancy-driven window ventilation (windows 50% open when the outside air is cooler and above 18 °C). The air conditioner keeps the same set point, schedules and COP as the fast engine. There are no internal gains (people, lamps, appliances) in either engine.

Package	Comfort % without A/C (EnergyPlus)	A/C kWh/yr fast	A/C kWh/yr EnergyPlus	Saving/yr EnergyPlus
Base	78.6	1,582	733	-
Rock wool 50 mm	84.8	952	265	R$ 468
Glass wool 50 mm	84.8	936	256	R$ 477
PET wool 50 mm	84.8	936	256	R$ 477
Sheep wool 50 mm	84.8	936	256	R$ 477
Rock wool 100 mm	85.1	854	208	R$ 525
Draught sealing	78.8	1,572	729	R$ 4
Double glazing	79.1	1,531	714	R$ 19
Low-E double glazing	79.2	1,409	668	R$ 65
Fixed brise 0.6 m	78.8	1,574	709	R$ 24
Movable blinds	78.6	1,573	698	R$ 35
All together	84.2	759	183	R$ 550

Result with EnergyPlus: base case: 78.6% of the hours in the comfort zone (fast engine 81.0%), 556 hot hours (fast 812), 1321 cold hours (fast 847); air-conditioning 733 kWh/yr (fast 1,582). Both engines agree on the ranking: **roof insulation is by far the best measure**, glazing and shading change little.

Differences between the engines are expected: EnergyPlus gives roughly half the air-conditioning energy of the fast engine and more cold hours (it couples the floor to the ground and loses heat through the attic more freely). For decisions, prefer the EnergyPlus figures; the fast engine remains useful for quick what-if runs. Not yet modelled: internal gains, door opening, the slope of the roof (a flat roof with the same attic volume is used) and the neighbours' shade.

8. Sustainable-building daylight classification (LEED / IES LM-83)

Calculated with: **Fast engine (Python, this project)** · Radiance cross-check in 8.1 · details and reliability in section 9

LEED is the most widely used green-building certification in the world. Its daylight credit (v4.1, Option 1) is awarded from the simulated **sDA300/50%** (share of the regularly occupied floor that gets 300 lux for at least half of the occupied hours) and the **ASE1000,250** (share of floor with more than 1,000 lux of direct sun for over 250 hours a year, a glare and overheating risk), both defined in the IES standard LM-83 (current edition LM-83-23, February 2023). Steps: average sDA of at least 40% = 1 point, 55% = 2 points, 75% = 3 points; an ASE above 10% must be justified with a glare strategy.

Room	Area m²	sDA300/50%	ASE1000,250	LEED step if alone
Living (front half)	21.2	2% No points 2/5	0% Very low 5/5	0 point(s)
TV / dining (rear half)	23.4	100% 3 points 5/5	0% Very low 5/5	3 point(s)
Bedroom 1	15.4	100% 3 points 5/5	10% High 2/5	3 point(s)
Bedroom 2	14.4	0% No points 2/5	0% Very low 5/5	0 point(s)
Bedroom 3	14.1	8% No points 2/5	0% Very low 5/5	0 point(s)
Kitchen	10.8	0% No points 2/5	0% Very low 5/5	0 point(s)
Whole house (area-weighted)	99.3	41% 1 point 3/5	2% Very low 5/5	1 point(s)

Result: area-weighted sDA300/50% = 40.6% and ASE1000,250 = 1.6%: **1 of 3 daylight point(s)**. Rooms below 40%: Living (front half), Bedroom 2, Bedroom 3, Kitchen.

This is an indicative pre-assessment: LEED requires a certified assessor, furniture and blind modelling, a 0.6 m grid and the full LM-83 procedure (our grid is 0.5 m, with 8:00-18:00 as occupied hours and blinds open). LEED v4.1 formally cites the 2012 edition of LM-83; the 2023 edition keeps the same two metrics (we could not verify the detailed changes). Homes are normally certified under LEED for Homes, whose daylight rules differ.

LEED daylight (sDA300/50%, area-weighted over the regularly occupied rooms) International standard

Average spatial daylight autonomy of the living rooms, bedrooms and kitchens (floor-area weighted): the share of floor that gets 300 lux for at least half of the occupied hours. LEED v4.1 gives 1 point from 40%, 2 points from 55% and 3 points from 75%. Indicative pre-assessment only: a real LEED credit needs a certified assessor and the full LM-83 method.

No points 2/5	below 40 % of floor	Below the 40% needed for the first point.
1 point 3/5	40 to 55 % of floor	Meets the first LEED daylight step.
2 points 4/5	55 to 75 % of floor	Meets the second LEED daylight step.
3 points 5/5	75 or more % of floor	Meets the top LEED daylight step.

This house: Whole house 41% 1 point 3/5

Source: USGBC LEED v4.1, Daylight credit, Option 1 (simulation): sDA300/50% average of 40% / 55% / 75% = 1 / 2 / 3 points; ASE1000,250 above 10% needs a glare strategy (thresholds read on the LEEDuser v4.1 credit page); ANSI/IES LM-83-23 (published Feb 2023): spatial daylight autonomy (sDA) and annual sunlight exposure (ASE); LEED v4.1 still cites the 2012 edition, same two metrics. Also needed: ASE1000,250 of at most 10% (otherwise explain the glare strategy). Wet rooms, halls and outdoor areas are not 'regularly occupied'.

8.1 Cross-check with a reference engine (Radiance)

Calculated with: **Radiance** · details and reliability in section 9

The numbers above come from our fast engine (ray-traced sun + a uniform-sky split-flux estimate). To test them, the same rooms and sensor points were recalculated with **Radiance** (Lawrence Berkeley National Laboratory), the open-source engine validated by CIE 171 and used by ClimateStudio and Honeybee: Perez all-weather sky from the same weather file, 4 light bounces, annual daylight-coefficient method, and the LM-83 rule that closes the blinds when direct sun hits more than 2% of the room (shade transmittance 5%). A sensor above the roof reproduces the weather file's illuminance within 1%, so the sky is calibrated.

Room	Area m²	sDA300/50% fast	sDA300/50% Radiance	ASE Radiance	Mean lux fast → Radiance
Living (front half)	21.2	2%	5% No points 2/5	0%	144 → 187
TV / dining (rear half)	23.4	100%	10% No points 2/5	7%	836 → 477
Bedroom 1	15.4	100%	15% No points 2/5	10%	998 → 706
Bedroom 2	14.4	0%	0% No points 2/5	0%	134 → 48
Bedroom 3	14.1	8%	0% No points 2/5	0%	213 → 119
Kitchen	10.8	0%	0% No points 2/5	0%	256 → 132

Result with Radiance: area-weighted sDA300/50% = 5.8% (fast engine: 40.6%), ASE1000,250 = 3.2%: **0 of 3 daylight point(s)**. The fast engine is optimistic deep inside the rooms; prefer the Radiance figures for decisions.

Not yet modelled in the Radiance run: furniture, frames and mullions (glass transmittance 0.80 is used for the whole opening), the glass-door lattice, the neighbours' buildings and the tree shade. Blinds follow the simplified per-room LM-83 rule. Windows and brise fixed as in the base case.

9. How this was calculated and how far to trust it

This section tells you how the numbers were produced, so you can judge how far to trust them. The fast engine is open-source Python code written for this project; it is cross-checked with three open-source reference engines: Radiance (daylight), EnergyPlus (heat, energy and NBR 15575) and OpenFOAM (wind pressure). The fast sub-models are standard in building physics but simplified. Nothing here has been measured on the real building.

> Validation status: the code is checked by regression tests (the base case must not change when a feature is added) and by sanity checks of sun geometry and energy balance; it has NOT been validated against measurements, and the reference engines were run on a simplified version of the model (see their limits). For a permit, certification or investment decision, confirm with a simulation by a qualified professional (e.g. EnergyPlus, Radiance).

Reference engines

RADSITE | radiance-online.org
Radiance 6.0.2
Daylight: sDA, ASE, LEED (LM-83)

EnergyPlus 26.2.0
Heat, energy and NBR 15575 (PHFT)

OpenFOAM®
OpenFOAM v2312 (Docker)
Wind CFD: pressure on the openings

Logos belong to their projects (Radiance: LBNL; EnergyPlus: US DOE / NREL; OpenFOAM: OpenCFD / ESI Group) and are shown only to credit the software used.

The engines in detail: parameters, formulas and references

Fast engine (Python, this project)

Used for: Sun and shadows, daylight (lux, lumens, sDA, ASE), window sizes and cross-ventilation hours, indoor temperature, comfort, air-conditioning / heating energy and cost, materials and upgrade scenarios (sections 2 to 7).

Parameters:

- 8,760 hourly steps of the typical weather year; sensor grid 0.5 m at 0.76 m height
- Daylight: luminous efficacy 100 lm/W (direct) and 115 lm/W (diffuse), mean reflectance 0.55, utilisation factor 0.45, maintenance factor 0.8
- Thermal: surface film resistances 0.04 (outside), 0.13 (wall), 0.17 (ceiling and floor) m^2K/W; outside film coefficient 17 W/m^2K; wall solar absorptance 0.45; glass U, g and VT from the catalogue
- Comfort: ASHRAE 55 adaptive band, 80% and 90% acceptability; running-mean outdoor temperature limited to 10-33.5 °C
- Cooling: set point and schedules per room type (project.json), COP and tariff as stated in section 5.3; windows closed while the equipment runs

Formulas:

- Lumens to install = lux × area ÷ (utilisation 0.45 × maintenance 0.8)
- Comfort temperature = 0.31 × T(outdoor running mean) + 17.8 °C; comfortable band = ± 3.5 °C (80%) or ± 2.5 °C (90%)
- Operative temperature = (air temperature + mean surface temperature) ÷ 2
- Electricity = heat removed ÷ COP; yearly cost = electricity × tariff
- Fixed brise: shadow depth = overhang × tan(profile angle); movable blinds: solar gain × 0.3 when the sun on the window exceeds 250 W/m^2
- sDA300/50% = share of the floor area with at least 300 lux for 50% of the occupied hours; ASE1000,250 = share of the area with 1,000 lux of direct sun for more than 250 hours

References:

- ASHRAE Standard 55: thermal environmental conditions for human occupancy (adaptive comfort model) - https://www.ashrae.org/technical-resources/standards-and-guidelines
- ABNT NBR 15575 (Residential buildings - performance) and NBR 15220 (Thermal performance of buildings) - https://www.abntcatalogo.com.br
- EN 17037: Daylight in buildings - https://standards.cencenelec.eu
- ANSI/IES LM-83-23: sDA and ASE - https://webstore.ansi.org/standards/iesna/ansiieslm8323
- Embree ray-tracing kernels (Intel) - https://github.com/embree/embree
- IfcOpenShell: open-source IFC toolkit - https://ifcopenshell.org
- pvlib python: a Python package for modeling solar energy systems (Holmgren, Hansen, Mikofski, JOSS 2018) - https://joss.theoj.org/papers/10.21105/joss.00884
- Numba: JIT compiler for Python - https://numba.pydata.org

EnergyPlus + AirflowNetwork

Used for: Cross-check of comfort and energy of the upgrade packages (section 7.4) and the whole NBR 15575-1:2021 simulation: PHFT, hottest and coldest operative temperature, thermal load, air changes through the openings (section 6.4).

Parameters:

- One thermal zone per room plus a ventilated attic zone; walls shared between rooms; conduction transfer functions for every layer
- Windows as simple glazing systems; real eaves as shading surfaces
- AirflowNetwork: every window, open door and crack is a flow path; wind pressure coefficients from the OpenFOAM CFD (24 directions), wind speed at the opening height from the power-law profile (exponent 0.22)
- NBR 15575 run: occupancy, lighting and equipment from the tables of the standard; windows open only while the room is occupied, the room air is at least 19 °C and warmer than outdoors; bathroom windows open
- Thermal-load run: windows closed, ideal air conditioner at 23 °C (cooling) and 21 °C (heating) while occupied
- Operative-temperature band 18-26 °C because the annual mean temperature of the weather file is below 25 °C
- Openings (SimpleOpening): discharge coefficient 0.6, opening factor = openable fraction of the window (0.5 sliding), closed-leak coefficient 0.00063 kg/s (windows) and 0.0024 kg/s (doors) at 1 Pa, exponent 0.63 / 0.59; attic eave openings 0.4% of the floor area (crack 0.0019 kg/s per m^2 of floor, exponent 0.65)
- Ventilation study (section 6.4): six full-year runs of the same model with floor-plan changes (doors closed, grille above the doors, casement windows 90% open, 0.2 m^2 ceiling grille per room with eave openings 5× larger, and the last two together) plus a run with the windows closed; the flow through each opening is read hour by hour (AFN Linkage Node Volume Flow Rate)
- Wind sectors of the flow arrows: eight sectors of 45° centred on N, NE, ... NW, using the wind direction of the weather file; hot hours = windows open and mean operative temperature of the bedrooms and living room above 26 °C

Formulas:

- PHFT = occupied hours with 18 °C ≤ operative temperature ≤ 26 °C ÷ occupied hours × 100
- Minimum level: PHFT(real) > 0.9 × PHFT(reference); T(max,real) ≤ T(max,ref) + 2 °C; T(min,real) ≥ T(min,ref) − 1 °C

- ΔPHFT = PHFT(real) − PHFT(reference); the intermediate and superior levels need ΔPHFT ≥ 45 − 0.58 × PHFT(reference)
- Load reduction = (load of the reference − load of the real house) ÷ load of the reference
- Airflow through an opening (AirflowNetwork, large vertical opening): driven by the pressure difference ΔP = wind pressure ($C_p \times \frac{1}{2}\rho U^2$) ± stack pressure ($\rho\, g\, \Delta h\, \Delta T/T$) between the two sides
- Air changes per hour (ACH) = air volume entering through the exterior openings in one hour ÷ room volume
- Flow through an opening: Q = Cd × A × opening factor × √(2 ΔP / ρ), with ΔP from the wind and stack pressures above; a crack follows Q = C × ΔP^n
- Heat carried out by ventilation = ρ cp Σ V(out) × (T(zone) − T(outdoor)), with ρ cp = 1.2 × 1005 J/(m³·K), summed over every exterior opening and ceiling-to-attic path, hour by hour (kWh per year)
- Exhaust effectiveness η = (T(closed) − T(ventilated)) ÷ (T(closed) − T(outdoor)), the operative temperatures of a room in the occupied hours when the closed house is above 26 °C and more than 1 °C above outdoors; 0 = ventilation does not cool, 1 = the room follows the outdoor air
- Degree-hours = Σ max(T(operative) − 26 °C, 0) over the occupied hours

References:

- EnergyPlus (US Department of Energy, NREL) - https://energyplus.net
- ANSI/ASHRAE Standard 140: Standard method of test for the evaluation of building energy analysis computer programs - https://www.ashrae.org/technical-resources/bookstore/standard-140
- EnergyPlus AirflowNetwork and surface-averaged wind pressure coefficients of Swami and Chandra (1987), EnergyPlus Engineering Reference - https://energyplus.net/documentation
- ABNT NBR 15575-1:2021 section 11.4 (thermal performance by simulation) and its public amendment draft (LabEEE, March 2021) - https://labeee.ufsc.br/sites/default/files/documents/P_ABNTNBR15575_1_2020CNGPR_PosCN_SiteLabEEE.pdf

Radiance

Used for: Daylight cross-check of the fast engine: sDA300/50%, ASE1000,250, mean lux and the LEED daylight points (section 8.1).

Parameters:

- Two-phase daylight coefficients (rfluxmtx), 4 light bounces
- Perez all-weather sky built from the weather file (gendaymtx), 2,305 sky patches, 8,760 hours
- Blind rule per room: blinds closed (5% transmittance) when more than 2% of the room gets 1,000 lux of direct sun
- Glass transmittance 0.80 for the whole opening; no furniture, frames, neighbours or trees

Formulas:

- Illuminance at a sensor = Σ (daylight coefficient × sky patch luminance), summed over the 2,305 patches for every hour
- sDA and ASE as defined by IES LM-83 (same definitions as the fast engine)

References:

- Radiance Synthetic Imaging System (Lawrence Berkeley National Laboratory) - https://www.radiance-online.org
- CIE 171:2006 Test cases to assess the accuracy of lighting computer programs - https://cie.co.at/publications/test-cases-assess-accuracy-lighting-computer-programs
- Perez et al. (1993): all-weather model for sky luminance distribution - https://doi.org/10.1016/0038-092X(93)90017-I
- ANSI/IES LM-83-23: sDA and ASE - https://webstore.ansi.org/standards/iesna/ansiieslm8323

OpenFOAM (CFD)

Used for: Wind pressure coefficients (Cp) in front of every window, door and eave opening, for 24 wind directions; they feed the EnergyPlus AirflowNetwork (section 6.4 and 6.5).

Parameters:

- Steady incompressible RANS (simpleFoam), standard k-epsilon turbulence model, wall functions
- Atmospheric boundary-layer inlet: 10 m/s at 10 m, roughness length z0 = 0.3 m (suburban)
- Domain 6H upstream, 15H downstream, 5H to each side, 6H above (H = building height); snappyHexMesh, 0.15 m cells on the house, coarser around the lot walls and neighbours
- Surroundings: lot walls and neighbouring buildings as closed boxes with heights estimated from photographs
- Convergence: residuals p 1e-5, U 1e-6, k and epsilon 1e-5; pressure is the mean of the last 20 iterations

Formulas:

- $C_p = p \div (\frac{1}{2}\, \rho\, U_EP(z)^2)$, with U_EP(z) = U10 × (z ÷ 10)^0.22 the wind profile EnergyPlus uses
- $Cp(\text{EnergyPlus}) = Cp(\text{CFD}) \times (U_CFD(z) \div U_EP(z))^2$, the correction for the different wind profiles (factor 0.39 to 0.79 with height)
- Reynolds-averaged momentum: $\nabla\cdot(U\,U) = -\nabla p/\rho + \nabla\cdot[(\nu + \nu_t)(\nabla U + \nabla U^T)]$, with $\nu_t = C\mu\, k^2 \div \varepsilon$ and $C\mu = 0.09$

References:

- OpenFOAM v2312 (OpenCFD / ESI Group), computational wind engineering with RANS k-epsilon - https://www.openfoam.com
- EN 1991-1-4: Eurocode 1, wind actions (table 7.1, pressure coefficients of vertical walls of rectangular buildings) - https://eurocodes.jrc.ec.europa.eu
- EnergyPlus AirflowNetwork and surface-averaged wind pressure coefficients of Swami and Chandra (1987), EnergyPlus Engineering Reference - https://energyplus.net/documentation

Software used

Python 3.11.9 · NumPy 2.4.6 · pandas 3.0.6 · pvlib 0.16.1 · trimesh 5.1.1 · Embree (embreex) 4.4.0 · IfcOpenShell 0.9.0 · Shapely 2.2.0 · Numba 0.68.0 · Matplotlib 3.11.2 · SciPy 1.17.1 · EnergyPlus 26.2.0 · Radiance 6.0.2 · OpenFOAM v2312 (Docker)

Calculation steps and how reliable each is

Step	How it is calculated	Reliability
3D model and orientation	The building is read from an IFC file (IfcOpenShell). Rooms are the IfcSpace volumes; walls, slabs, roofs, windows and doors become triangle meshes. The model is rotated so that true north comes from the IFC True North (plus an optional correction). Why: Exact as far as the model is exact: sizes, window positions and materials are estimates unless the model comes from a survey.	High
Sun position and shadows	Solar position for each of the 8,760 hours with pvlib (NREL SPA-based algorithm). From a grid of sensor points (0.5 m, at 0.76 m height) in every room a ray is traced towards the sun against all opaque elements with Embree (a ray-tracing library): blocked = shadow; a ray through a window is attenuated by the window transmittance. Why: Sun geometry is deterministic; the result is only as good as the geometry (neighbours and trees count only if they are in the model).	High
Weather	A typical meteorological year of hourly data (EnergyPlus .epw such as TMYx from climate.onebuilding.org, or a typical year from PVGIS): temperature, humidity, direct/diffuse/global radiation, wind speed and direction. Why: A typical year is not a forecast: real years are hotter or colder. Wind data are for open terrain at 10 m.	Medium
Daylight (lux and lumens)	Illuminance on the work plane = direct sun (ray traced) + sky light (isotropic sky, sky-view factor by ray tracing) + an inter-reflected component (split-flux method, mean reflectance 0.55). Radiation is converted to lux with luminous efficacies of 100 lm/W (direct) and 115 lm/W (diffuse). Lamp lumens use the lumen method: lux x area / (utilisation 0.45 x maintenance 0.8). Metrics: NBR 15575 centre-of-room lux, sDA300/50% and ASE1000,250 (IES LM-83), EN 17037 sunlight hours. Why: The isotropic sky ignores the brighter sky near the sun and split-flux overestimates light deep in the rooms: the Radiance cross-check gives an area-weighted sDA of 6% against 41% here. Use it only for quick comparisons.	Medium
Reference engine for daylight: Radiance	The same rooms and sensor points are recalculated with Radiance (Lawrence Berkeley National Laboratory), a backward ray tracer validated against the CIE 171:2006 test cases and the engine behind ClimateStudio and Honeybee. Method: two-phase daylight coefficients (rfluxmtx, 4 light bounces) multiplied by a Perez all-weather sky built from the weather file (gendaymtx, 2,305 patches) for all 8,760 hours; sDA300/50% and ASE1000,250 as in IES LM-83, with the blind rule approximated per room (blinds closed, 5% transmittance, when more than 2% of the room gets 1,000 lux of direct sun). A sensor above the roof checks the sky: it reproduces the weather file's illuminance within about 2%. Why: Validated engine; what is still missing in the run is furniture, frames, the glass-door lattice, neighbours and trees, and the blind rule is simplified. Prefer these numbers over the fast daylight engine.	High
Reference engine for heat and energy: EnergyPlus	The retrofit packages are recalculated with EnergyPlus (US Department of Energy), the heat-balance engine validated by ASHRAE Standard 140 and used inside DesignBuilder and OpenStudio. One zone per room plus a ventilated attic zone above it; walls shared between rooms; conduction transfer functions for every layer; windows as simple glazing systems; the real eaves as shading surfaces; window ventilation by wind and buoyancy (windows 50% open when the outside air is cooler and above 18 C); an ideal-loads air conditioner with the same set point, schedules and COP as the fast engine. Comfort comes from a free-floating run (ASHRAE 55 adaptive), energy from a second run with the air conditioner. Why: Validated engine, but the model is simplified: flat roof with the same attic volume, no internal gains, no door opening, no neighbours' shade, window glass properties assumed, not calibrated with measurements. Compare options; do not treat the bill as a forecast.	Medium
NBR 15575-1:2021 by simulation and cross ventilation (EnergyPlus + AirflowNetwork)	A real model and a reference model of the house are simulated for the 8,760 hours with EnergyPlus. Natural ventilation is solved by the AirflowNetwork: windows, open doors and cracks are flow paths driven by wind pressure (Swami and Chandra facade coefficients for a low-rise building, local wind speed at the opening height), stack effect and opening height; windows open only while the room is occupied, when the room is at least 19 C and warmer than outdoors; bathroom windows stay open. Occupancy, lighting and equipment follow the standard's tables; a second run with windows closed and an ideal air conditioner (23 C cooling / 21 C heating while occupied) gives the thermal load. PHFT, hottest and coldest operative temperature and load reduction give the minimum, intermediate and superior levels. Why: Follows section 11.4 of NBR 15575-1:2021 (full text, tables checked against the public amendment draft); the bioclimatic zone of Maringa in NBR 15220-3:2024 comes from a non-official article; our assumptions are listed in the report (transient rooms, open doors, ground temperature). The wind pressure coefficients come from the OpenFOAM CFD of this house with its lot walls and neighbours (heights estimated), not from a wind-tunnel test. Not an official compliance report.	Medium
Air circulation, floor-plan changes and hot-air exhaust (EnergyPlus + AirflowNetwork)	The same airflow network is run for the whole year with several floor-plan changes (doors closed, grille above the doors, casement windows, ceiling exhaust). The flow through each window, door, grille and ceiling opening is read hour by hour and averaged for the hours with open windows, the hot hours and each wind direction, which gives the arrows on the plan. Exhaust effectiveness compares the operative temperature of each room with a run with the windows closed and with the outdoor temperature; the heat carried out by the air is rho cp V (T zone - T outdoor). Why: The flows are averages between rooms (one air node per room, no velocity field and no air-age or stagnant-zone calculation); the wind pressure comes from the CFD of the outside only, with closed openings, and the grille, ceiling opening and casement fraction are idealised sizes, not products. Good for ranking plan changes (differences between variants are more reliable than absolute flows); not a replacement for an interior CFD or a smoke test.	Medium
Wind pressure on the facades: OpenFOAM (CFD)	For each of 24 wind directions (every 15 degrees) the house (a closed envelope rebuilt from the IFC roofs, eaves and porches) is placed in a computational domain (6H upstream, 15H downstream, 5H to the sides, 6H above, H = building height) and meshed with snappyHexMesh (1.2 to 2.0 million cells, 0.15 m on the house, coarser around the lot walls and neighbours). OpenFOAM v2312 solves the steady Reynolds-averaged Navier-Stokes equations with the standard k-epsilon turbulence model, an atmospheric-boundary-layer inlet (10 m/s at 10 m, roughness length 0.3 m) and rough-ground wall functions. The lot walls and the neighbouring buildings are included as closed boxes (heights estimated from photographs). The pressure 0.3 m in front of every window, door and eave opening, converted to a coefficient of the dynamic pressure of the EnergyPlus wind profile (power law, suburban terrain) at that height, gives the wind pressure coefficients handed to the EnergyPlus AirflowNetwork. A numerical verification (three meshes, roughness, domain size, a box compared with the Eurocode) gives the uncertainty of each coefficient. It runs in a Docker container. Why: RANS k-epsilon is the usual engineering model for mean wind loads on buildings and OpenFOAM is validated for it, but this run uses the real roof shape, lot walls and neighbours whose heights are estimates from photographs, no trees, a single approach roughness and a mesh of moderate size; no wind-tunnel check was made.	Medium
Indoor temperature (thermal model)	Each room is a lumped heat-balance model solved hour by hour (Python, compiled with Numba): wall, ceiling and floor layers as heat-storage nodes (resistances and capacities from the layer conductivity, density, specific heat and thickness, ISO 6946 / NBR 15220-2 style), sol-air temperature on the outside surfaces, a ventilated attic, windows (U-value, solar gain), ventilation (infiltration when closed, cross-ventilation by wind when open, rooms linked through doors) and thermal mass. Operative temperature = mean of air and surface temperature. Comfort uses the ASHRAE 55 adaptive model (80% and 90% acceptability) and the NBR 15575 hottest/coldest-day checks. Why: A simplified one-zone-per-room model, not calibrated with measurements. It is cross-checked against EnergyPlus for the retrofit packages (see the EnergyPlus step): the engines agree on the ranking of measures but EnergyPlus gives about half the air-conditioning energy. Use it for quick comparisons, not as an exact forecast.	Medium
Air-conditioning / heating energy and cost	When the room is above the cooling set point (or below the heating set point) during its schedule, the model adds the heat flow needed to hold the set point; electricity = heat / COP (cooling) or fuel = heat / efficiency (heating), cost = energy x tariff. Windows are closed while the equipment runs. Sensible heat only; no people, appliances or lighting heat. Why: Real bills differ: latent load, occupants' habits, equipment efficiency and the tariff (an assumption) all matter. Treat the cost as an order of magnitude and as a way to compare options.	Low
Wind and cross-ventilation	Wind roses from the weather file; each window is classified windward / leeward / side-on for each hour; rooms whose connected windows face both the wind and the lee are given more air changes. No pressure calculation (the CFD and the AirflowNetwork are used only in section 6.4). Why: A rule-based estimate: wind at the house is lower and more turbulent than at the weather station; no pressure-driven flow calculation.	Low
Upgrade scenarios	Each upgrade changes the base model and the whole year is simulated again: roof insulation adds a conduction layer (thickness / conductivity) on the attic side of the ceiling; draught sealing lowers the closed-house air change rate; glazing changes the window U-value, solar gain factor and visible transmittance; a fixed brise cuts direct sun by the shadow depth = overhang x tan(profile angle); movable blinds multiply the solar gain by 0.3 whenever the sun on the window is above 250 W/m2 during the day. Why: Good for ranking options (differences between scenarios are more reliable than absolute values). Material properties are catalogue values; their confidence is shown next to each value. Shading is analytic, not ray traced.	Medium
Ratings and LEED pre-assessment	Numbers are compared with ABNT norms where they exist (NBR 15575, NBR 15220, NBR 5413, NBR 16401), international standards otherwise (EN 17037, IES LM-83, ASHRAE 55, WHO) and a few rules of thumb that are labelled as such. The LEED v4.1 daylight points follow Option 1 (sDA300/50% and ASE1000,250). Why: Several norm values were read from public summaries or amendment drafts, not the full ABNT texts; the LEED result is an indicative pre-assessment, not a certification.	Medium

References

- pvlib python: a Python package for modeling solar energy systems (Holmgren, Hansen, Mikofski, JOSS 2018) - https://joss.theoj.org/papers/10.21105/joss.00884
- PVGIS typical meteorological year (European Commission JRC) - https://re.jrc.ec.europa.eu/pvg_tools/en/
- Climate.OneBuilding.Org: TMYx weather files - https://climate.onebuilding.org
- Embree ray-tracing kernels (Intel) - https://github.com/embree/embree
- IfcOpenShell: open-source IFC toolkit - https://ifcopenshell.org
- Numba: JIT compiler for Python - https://numba.pydata.org
- ASHRAE Standard 55: thermal environmental conditions for human occupancy (adaptive comfort model) - https://www.ashrae.org/technical-resources/standards-and-guidelines
- ABNT NBR 15575 (Residential buildings - performance) and NBR 15220 (Thermal performance of buildings) - https://www.abntcatalogo.com.br
- EN 17037: Daylight in buildings - https://standards.cencenelec.eu
- Radiance Synthetic Imaging System (Lawrence Berkeley National Laboratory) - https://www.radiance-online.org
- CIE 171:2006 Test cases to assess the accuracy of lighting computer programs - https://cie.co.at/publications/test-cases-assess-accuracy-lighting-computer-programs
- EnergyPlus (US Department of Energy, NREL) - https://energyplus.net
- ANSI/ASHRAE Standard 140: Standard method of test for the evaluation of building energy analysis computer programs - https://www.ashrae.org/technical-resources/bookstore/standard-140
- Perez et al. (1993): all-weather model for sky luminance distribution - https://doi.org/10.1016/0038-092X(93)90017-I
- EnergyPlus AirflowNetwork and surface-averaged wind pressure coefficients of Swami and Chandra (1987), EnergyPlus Engineering Reference - https://energyplus.net/documentation
- ABNT NBR 15575-1:2021 section 11.4 (thermal performance by simulation) and its public amendment draft (LabEEE, March 2021) - https://labeee.ufsc.br/sites/default/files/documents/P_ABNTNBR15575_1_2020CNGPR_PosCN_SiteLabEEE.pdf
- OpenFOAM v2312 (OpenCFD / ESI Group), computational wind engineering with RANS k-epsilon - https://www.openfoam.com
- ANSI/IES LM-83-23: sDA and ASE - https://webstore.ansi.org/standards/iesna/ansiieslm8323
- USGBC LEED v4.1 Daylight credit (Option 1: simulation) - https://leeduser.buildinggreen.com/credit/CI-v4.1/EQc7
- AIVC: air infiltration reduction by weatherstripping retrofits - https://www.aivc.org/resource/reduction-air-infiltration-due-window-and-door-retrofits-older-home
- WHO Housing and Health Guidelines (2018) - https://www.who.int/publications/i/item/9789241550376
- EN 1991-1-4: Eurocode 1, wind actions (table 7.1, pressure coefficients of vertical walls of rectangular buildings) - https://eurocodes.jrc.ec.europa.eu

10. Assumptions, limits and sources

- **Dimensions** come from the 3D model supplied: a video scan plus the listing; this is not a survey.
- **Materials**: the standard Brazilian 8-hole ceramic block with plaster and a clay-tile roof with concrete slab ceiling, not confirmed for this house. Neighbouring buildings, street trees and curtains are not modelled unless they are in the model.
- **Doors** are assumed open for light and air. Furniture, people, appliances and lighting heat are not included (empty house).
- The comfort model is a simplified hourly heat balance, not calibrated against measurements: use it to compare rooms and options. EnergyPlus would be the next level of rigour.
- **Wind** data are for open terrain at 10 m; inside a plot with walls and neighbours the speeds are lower. The fast engine's wind rule ignores this; the pressure-based ventilation of section 6.4 includes the walls and neighbours through the CFD, with estimated heights.
- **Daylight** uses an isotropic sky; real partly-cloudy skies are brighter near the sun.
- Rating scales marked *Rule of thumb* are ours; the ones marked norm come from the documents listed below. Several norm values were read from public summaries and amendment drafts: confirm them in the full ABNT texts before using them in a legal, contractual or financing decision.

Sources

- ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements
- ABNT NBR 15575-1:2021 + amendment (thermal performance by simulation vs. a reference house)
- ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas
- ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs
- ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3)
- ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels)
- ABNT NBR ISO/CIE 8995-1:2013 (workplace lighting; reading/writing tasks 300-500 lx)
- EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h)
- IES LM-83-12 (sDA and ASE), as used by LEED v4.1
- ASHRAE 55 adaptive comfort model (80% / 90% acceptability)
- ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH
- Passive House Institute criterion: heating demand of at most 15 kWh per m2 per year
- USGBC LEED v4.1, Daylight credit, Option 1 (simulation): sDA300/50% average of 40% / 55% / 75% = 1 / 2 / 3 points; ASE1000,250 above 10% needs a glare strategy (thresholds read on the LEEDuser v4.1 credit page)
- ANSI/IES LM-83-23 (published Feb 2023): spatial daylight autonomy (sDA) and annual sunlight exposure (ASE); LEED v4.1 still cites the 2012 edition, same two metrics
- WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C)
- WMO Beaufort wind scale
- Rule of thumb of this study (no norm gives a number)

Run it again after changing the model

Edit the 3D model (house.ifc), then double-click **run_all.bat** in the House3D folder. It redoes the simulation, this PDF and the data for the 3D viewer (viewer / open_viewer.bat).